LaTeX Beginner Handbook

A Beginner Handbook for Coursework and Academic Paper Writing

Who this is for. You may already know your field and its research methods, but know almost nothing about LaTeX. This handbook starts from zero and focuses on the commands and habits you are most likely to use in problem sets, course notes, research drafts, and academic papers. Many worked examples come from economics and quantitative social science, but the LaTeX techniques are intended for students and researchers across fields.

How to use it. Read Parts I–III once. After that, use the later sections as a lookup reference. You do *not* need to memorize the commands. Copy a working example, change one thing, compile, and repeat.

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Part I

Start Here: How LaTeX Works

1 The One-Minute Mental Model

LaTeX is a writing system in which you type plain text plus instructions. You save those instructions in a file ending in `.tex`. A LaTeX engine reads the file and creates a PDF.

What you type

```
The parameter of interest is $\beta$.
```

What appears in the PDF

The parameter of interest is β.

The command-like parts are not supposed to appear literally in the PDF. They tell LaTeX what to typeset.

A useful mental model is:

`main.tex` ⟶ **compile** ⟶ `main.pdf`

Beginner tip. Your source file is not the final product. The PDF is the final typeset result.

2 Your First File: The Smallest Working Document

What you type

```
\documentclass[11pt]{article}

\begin{document}

Hello, LaTeX.

\end{document}
```

What appears in the PDF

Hello, LaTeX.

What it means

- `\documentclass[11pt]{article}` chooses a general article layout and an 11-point base font.
- `\begin{document}` marks the start of material that will appear in the PDF.
- `\end{document}` marks the end.

Everything before `\begin{document}` is called the **preamble**. Packages, custom commands, title information, and document settings normally go there.

3 What Does "Compile" Mean?

Compiling means asking LaTeX to turn the `.tex` source into a PDF. In Overleaf you press **Recompile**. In a local editor such as VS Code, TeXstudio, or RStudio, you run a LaTeX build command.

A simple project might contain:

```
main.tex your paper source
main.pdf compiled paper
main.aux temporary cross-reference information
main.log compiler log
main.out hyperlink information
references.bib bibliography database (if used)
```

You normally edit `main.tex`. The auxiliary files are generated automatically.

3.1 Why do I sometimes need to compile twice?

LaTeX may need one pass to discover numbers and a second pass to insert them. This is common for:

- section, equation, figure, and table references;
- table of contents entries;
- citations and bibliographies.

If a reference appears as **??**, compile again before assuming something is wrong.

4 Commands, Arguments, Options, and Environments

Most LaTeX syntax falls into four patterns.

4.1 A command

What you type

```
\LaTeX
```

What appears in the PDF

LaTeX

A command begins with a backslash.

4.2 A command with a required argument

What you type

```
\textbf{Important result}
```

What appears in the PDF

Important result

The braces `{...}` contain the material the command acts on.

4.3 A command with an optional argument

What you type

```
\documentclass[11pt]{article}
```

What appears in the PDF

This line does not print body text by itself. It selects the `article` class and an 11-point base font; the visible document uses those settings.

Square brackets usually contain optional settings; braces usually contain required information.

4.4 An environment

What you type

```
\begin{itemize}
  \item First point
  \item Second point
\end{itemize}
```

What appears in the PDF

- First point
- Second point

Every `\begin{name}` must eventually have a matching `\end{name}`.

5 Packages: Adding Features

Packages add functionality. A practical beginner preamble for academic paper writing is:

```
\documentclass[11pt]{article}

\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb}
\usepackage{graphicx}
\usepackage{subcaption}
\usepackage{pdflscape}
\usepackage{booktabs}
\usepackage{threeparttable}
\usepackage[round,authoryear]{natbib}
\usepackage[hidelinks]{hyperref}
```

Package	Why you may want it
`geometry`	Set margins simply.
`amsmath`	Better equations and environments such as `align`.
`amssymb`	Extra mathematical symbols.
`graphicx`	Insert PDF/PNG/JPG figures.
`booktabs`	Professional horizontal rules for tables.
`threeparttable`	Clean notes below tables.
`natbib`	Author-year citations such as "Author (2020)" and "(Author, 2020)."
`hyperref`	Clickable cross-references and links.

Beginner tip. Start with a small preamble. Add packages when you actually need them. A huge preamble makes debugging harder.

Part II

Text, Punctuation, and Document Structure

6 Plain Text, Paragraphs, and Line Breaks

You type ordinary prose normally.

What you type

```
This paper studies how households respond to a change in prices.
```

What appears in the PDF

This paper studies how households respond to a change in prices.

To create a new paragraph, leave a blank line in the source.

What you type

```
This is the first paragraph.

This is the second paragraph.
```

What appears in the PDF

This is the first paragraph.

This is the second paragraph.

Common mistake. Do not use repeated \\ commands to create ordinary paragraph spacing. A blank source line is the normal paragraph break.

7 Bold, Italic, Emphasis, and Typewriter Text

Purpose	Source	Compiled result
Bold	`\textbf{important}`	**important**
Italic	`\textit{term}`	*term*
Emphasis	`\emph{key idea}`	*key idea*
Code/file name	`\texttt{main.tex}`	`main.tex`

Beginner tip. In academic papers, use bold and italics sparingly. Section structure should do most of the organizational work.

8 Reserved Characters: Characters LaTeX Treats Specially

Some keyboard characters are instructions to LaTeX. If you want the literal character in prose, you must escape or replace it.

Character	Type this to print it	Why it is special
%	`\%`	Starts a comment.
$	`\$`	Starts/ends inline math mode.
&	`\&`	Separates columns/alignment points.
_	`\_`	Creates a subscript in math mode.
#	`\#`	Used for macro parameters.
{	`\{`	Starts a grouped argument.
}	`\}`	Ends a grouped argument.
~	`\textasciitilde`	~ is an accent command; bare is a nonbreaking space.
^	`\textasciicircum`	Superscript operator in math mode.
\	`\textbackslash`	Starts most LaTeX commands.

What you type

```
R\&D spending increased by 5\%.
The replication file is data\_clean.csv.
The fee is \$25.
See Figure \#2.
```

What appears in the PDF

R&D spending increased by 5%. The replication file is data_clean.csv. The fee is $25. See Figure #2.

Common mistake. Typing `5%` without the backslash makes LaTeX treat the rest of that source line as a comment.

9 Comments: Notes That Do Not Print

Anything after an unescaped percent sign is ignored until the end of that source line.

What you type

```
% TODO: rewrite this paragraph.
```

```
This sentence appears in the PDF. % private note to myself
```

What appears in the PDF

This sentence appears in the PDF.

Comments are useful for TODOs, temporarily disabling text, and organizing long source files.

10 Quotation Marks, Dashes, and Small Typography Habits

Traditional LaTeX quotation marks are typed with two backticks and two apostrophes.

What you type

```
``Identification requires a stronger assumption.''
```

What appears in the PDF

“Identification requires a stronger assumption.”

Use different numbers of hyphens for different dashes:

Source	Result	Typical use
`1-5`	1-5	hyphen / simple connection
`1990-2000`	1990–2000	numerical range (en dash)
`word--word`	word—word	parenthetical break (em dash)

For references, a tilde creates a nonbreaking space:

What you type

```
Table~\ref{tab:main}
Figure~\ref{fig:main}
Section~\ref{sec:data}
```

What appears in the PDF

For example, if the referenced objects are numbered 1, 2, and 3:

Table 1 Figure 2 Section 3.

The actual numbers are supplied automatically from the corresponding labels.

This keeps “Table” and its number on the same line.

11 Titles, Authors, Dates, and Abstracts

What you type

```
\title{The Effect of X on Y}
\author{Your Name\\Your University}
\date{September 2026}
```

```
\begin{document}
\maketitle

\begin{abstract}
This paper studies ...
\end{abstract}
```

What appears in the PDF

The Effect of X on Y

Your Name

Your University

September 2026

Abstract

This paper studies ...

If you do not want a date, use `\date{}`.

12 Sections and Subsections

What you type

```
\section{Introduction}
\subsection{Data Sources}
\subsubsection{Sample Construction}
```

What appears in the PDF

1 Introduction

1.1 Data Sources

1.1.1 Sample Construction

The exact numbers depend on where these headings occur in the document.

LaTeX handles the numbering automatically. Do not type "1. Introduction" yourself.

For an unnumbered heading:

```
\section*{Acknowledgments}
```

13 A Beginner-Friendly Research Paper Structure

A common empirical-paper skeleton is:

```
\section{Introduction}
\section{Background or Institutional Setting}
\section{Data}
\section{Research Design or Empirical Strategy}
\section{Results}
\section{Robustness and Extensions}
\section{Conclusion}
```

The exact headings depend on the paper. LaTeX does not determine your research structure; it only typesets it consistently.

Part III

Math from Zero: Syntax You Will Reuse

14 Math Mode: The Most Important Distinction

LaTeX treats text and mathematics differently.

14.1 Inline math

What you type

```
The coefficient of interest is $\beta$.
```

What appears in the PDF

The coefficient of interest is β.

Use inline math for short expressions inside a sentence.

14.2 Displayed math

What you type

```
\[
y_i = \alpha + \beta x_i + \varepsilon_i.
\]
```

What appears in the PDF

$$y_i = \alpha + \beta x_i + \varepsilon_i.$$

Common mistake. Do not write math commands such as `\beta` in ordinary text mode. Write `$\beta$`.

15 Subscripts and Superscripts

What you type

```
$x_i$
$x_{it}$
$x^2$
$x^{10}$
$x_{it}^2$
```

What appears in the PDF

x_i, x_{it}, x^2, x^{10}, x_{it}^2.

Use braces whenever the subscript or superscript contains more than one token.

Common mistake. `x_it` does not mean x_{it}. Write `x_{it}`.

16 Greek Letters

Source	Result	Source	Result
`\alpha`	α	`\beta`	β
`\gamma`	γ	`\delta`	δ
`\epsilon`	ϵ	`\varepsilon`	ε
`\theta`	θ	`\lambda`	λ
`\mu`	μ	`\pi`	π
`\rho`	ρ	`\sigma`	σ
`\phi`	ϕ	`\omega`	ω
`\Delta`	Δ	`\Sigma`	Σ

17 Fractions, Roots, Parentheses, and Brackets

What you type

```
$\frac{a}{b}$
$\sqrt{x}$
$\sqrt[3]{x}$
```

What appears in the PDF

$\frac{a}{b}$, $\sqrt{x}$, $\sqrt[3]{x}$.

For tall expressions, use automatically sized delimiters:

What you type

```
\[
\left( \frac{a+b}{c} \right)
\qquad
\left[ \frac{a+b}{c} \right]
\]
```

What appears in the PDF

$$\left(\frac{a+b}{c}\right) \qquad \left[\frac{a+b}{c}\right].$$

Literal curly braces inside math need backslashes:

What you type

```
\[
\left\{x \in \mathbb{R}: x>0\right\}
\]
```

What appears in the PDF

$$\left\{x \in \mathbb{R} : x > 0\right\}.$$

18 Common Mathematical Symbols

Source	Result	Meaning	Source	Result	Meaning
`\le`	$\le$	less/equal	`\ge`	$\ge$	greater/equal
`\neq`	$\neq$	not equal	`\approx`	$\approx$	approximately
`\in`	$\in$	belongs to	`\notin`	$\notin$	not in
`\infty`	∞	infinity	`\pm`	$\pm$	plus/minus
`\to`	$\to$	tends/goes to	`\implies`	$\implies$	implies
`\times`	$\times$	multiplication	`\cdot`	$\cdot$	centered dot

19 Sums, Products, Integrals, and Limits

What you type

```
\[
\sum_{i=1}^{n} x_i
\qquad
\prod_{i=1}^{n} x_i
\]

\[
\int_a^b f(x)\,\mathrm{d}x
\qquad
\lim_{n\to\infty} a_n
\]
```

What appears in the PDF

$$\sum_{i=1}^{n} x_i \qquad \prod_{i=1}^{n} x_i$$

$$\int_a^b f(x)\,\mathrm{d}x \qquad \lim_{n\to\infty} a_n.$$

The pattern is reusable: `_{lower}` puts something below, and `^{upper}` puts something above.

20 Derivatives and Partial Derivatives

What you type

```
\[
\frac{\mathrm{d}y}{\mathrm{d}x}
\qquad
\frac{\partial f}{\partial x}
\qquad
\frac{\partial^2 f}{\partial x^2}
\]
```

What appears in the PDF

$$\frac{\mathrm{d}y}{\mathrm{d}x} \qquad \frac{\partial f}{\partial x} \qquad \frac{\partial^2 f}{\partial x^2}.$$

21 Math Operators: Use Commands Instead of Italic Letters

Commands such as `\log`, `\ln`, `\exp`, `\max`, and `\min` give proper operator spacing.

What you type

```
\[
\log x,\quad \ln x,\quad \exp(x),\quad
\max_x f(x)
\]
```

What appears in the PDF

$$\log x, \quad \ln x, \quad \exp(x), \quad \max_x f(x).$$

Common mistake. Writing `max` directly in math produces italic letters max, not the mathematical operator max.

22 Expectations, Probability, Variance, and Covariance

What you type

```
\[
\mathbb{E}[Y\mid X]
\qquad
\mathbb{P}(A)
\qquad
```

```
\operatorname{Var}(X)
\qquad
\operatorname{Cov}(X,Y)
\]
```

What appears in the PDF

$$\mathbb{E}[Y \mid X] \qquad \mathbb{P}(A) \qquad \operatorname{Var}(X) \qquad \operatorname{Cov}(X,Y).$$

If you use these constantly, define shortcuts in the preamble:

```
\newcommand{\E}{\mathbb{E}}
\newcommand{\Prob}{\mathbb{P}}
\newcommand{\Var}{\operatorname{Var}}
\newcommand{\Cov}{\operatorname{Cov}}
```

Then `\E[Y\mid X]` produces $\mathbb{E}[Y \mid X]$.

23 Hats, Bars, Tildes, Stars, and Bold Symbols

Source	Result	Typical reading	Comment
`\hat{\beta}`	$\hat{\beta}$	beta hat	estimate or fitted quantity
`\bar{x}`	$\bar{x}$	x bar	average / transformed notation
`\tilde{x}`	$\tilde{x}$	x tilde	transformed/alternative variable
`x\^*`	x^*	x star	optimum/equilibrium/etc.
`\mathbf{x}`	$\mathbf{x}$	bold x	vector/matrix notation
`\bm{\beta}`	$\boldsymbol{\beta}$	bold beta	bold Greek symbol

24 Matrices and Vectors

What you type

```
\[
\mathbf{x}=
\begin{pmatrix}
x_1\\
x_2\\
x_3
\end{pmatrix}
\qquad
A=
\begin{pmatrix}
a_{11} & a_{12}\\
a_{21} & a_{22}
\end{pmatrix}
\]
```

What appears in the PDF

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Inside matrices, `&` separates columns and `\\` ends a row.

25 Cases and Text Inside Math

What you type

```
\[
D_i=
\begin{cases}
1, & \text{if unit $i$ is treated},\\
0, & \text{otherwise}.
\end{cases}
\]
```

What appears in the PDF

$$D_i = \begin{cases} 1, & \text{if unit } i \text{ is treated}, \\ 0, & \text{otherwise}. \end{cases}$$

Use `\text{...}` for ordinary words inside math mode.

Part IV

Equations in an Academic Paper

26 Unnumbered vs. Numbered Equations

Use `\[ ... \]` when you do not need to refer to the equation later. Use the `equation` environment when the equation deserves a number.

What you type

```
\begin{equation}
y_i = \alpha + \beta x_i + \varepsilon_i.
\label{eq:baseline}
\end{equation}
```

What appears in the PDF

$$y_i = \alpha + \beta x_i + \varepsilon_i. \tag{1}$$

Then write:

What you type

```
Equation~\eqref{eq:baseline} defines the baseline specification.
```

What appears in the PDF

Equation (1) defines the baseline specification.

27 Multi-Line Equations with align

The `align` environment is one of the most useful tools in technical academic writing.

What you type

```
\begin{align}
A
&= B+C,\\
&= D+E,\\
&= F.
\end{align}
```

What appears in the PDF

$$A = B + C, \tag{2}$$
$$= D + E, \tag{3}$$
$$= F. \tag{4}$$

The `&` marks the alignment point. Usually you place it before the equals sign.

For no equation numbers, use `align*`.

Common mistake. Inside `align`, a literal ampersand is not "and." It controls alignment.

28 Long Equations: split and aligned

If one numbered equation is too wide, split it over multiple lines while keeping one equation number.

What you type

```
\begin{equation}
\begin{split}
y_i
&= \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} \\
&\quad + \beta_3 x_{3i} + \varepsilon_i.
\end{split}
\label{eq:long}
\end{equation}
```

What appears in the PDF

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i. \tag{5}$$

29 Equation Punctuation: Equations Are Part of Sentences

A displayed equation usually belongs grammatically to the sentence around it. Punctuate it accordingly.

What you type

```
We define the transformed outcome as
\[
\tilde y_i = y_i-\bar y,
\]
where $\bar y$ denotes the sample mean.
```

What appears in the PDF

We define the transformed outcome as

$$\tilde{y}_i = y_i - \bar{y},$$

where $\bar{y}$ denotes the sample mean.

Notice the comma after the equation because the sentence continues.

30 Explain Notation After an Equation

Do not display a complicated equation and leave the reader to infer every symbol.

What you type

```
We write the estimating equation as
\begin{equation}
y_{it}=\alpha+\beta d_{it}+\mathbf{x}_{it}'\gamma+\varepsilon_{it}.
\label{eq:model}
\end{equation}
Here, $y_{it}$ is the outcome, $d_{it}$ is the variable of interest,
$\mathbf{x}_{it}$ is a vector of controls, and $\varepsilon_{it}$ is an error term.
```

What appears in the PDF

We write the estimating equation as

$$y_{it} = \alpha + \beta d_{it} + \mathbf{x}_{it}'\gamma + \varepsilon_{it}.$$

Here, y_{it} is the outcome, d_{it} is the variable of interest, $\mathbf{x}_{it}$ is a vector of controls, and ε_{it} is an error term.

Beginner tip. This is a writing habit, not just a LaTeX habit: define notation when the reader first needs it.

31 Useful Spacing Commands in Math

Most spacing should be automatic, but a few commands are useful:

Command	Size	Example
`\,`	small space	$f(x)\,dx$
`\quad`	larger space	$A \quad B$
`\qquad`	even larger	$A \qquad B$
`\!`	small negative space	occasionally tightens notation

Do not manually space every operator. Let LaTeX do most of the work.

Part V

Small Commands and Layout Tricks You Will Use Constantly

32 Why These "Tiny" Commands Matter

Once you know basic LaTeX, much of the day-to-day improvement in a paper comes from small commands rather than new environments. Commands such as `\qquad`, `~`, `\hfill`, `\noindent`, and `\notag` help you control spacing, alignment, line breaking, references, and mathematical typography without manually fighting the page.

A useful way to remember them is by job:

Job	Commands worth recognizing
Small math spacing	`\,`, `\:`, `\;`, `\!`, `\quad`, `\qquad`
Horizontal layout	`\hspace`, `\hfill`, `\makebox`, `\phantom`
Vertical layout	`\smallskip`, `\medskip`, `\bigskip`, `\vspace`
Paragraphs and alignment	blank line, `\par`, `\noindent`, `\centering`, `\raggedright`
Equation alignment	`&`, `\\`, `\notag`, `\intertext`, `aligned`
Math typography	`\text`, `\mathrm`, `\mathbf`, `\bm`, `\mathbb`, `\operatorname`
Delimiter sizing	`\left`, `\right`, `\bigl`, `\Bigr`
References and links	`~`, `\ref`, `\eqref`, `\pageref`, `\autoref`, `\href`
Page control	`\newpage`, `\clearpage`, `\pagebreak`, `\FloatBarrier`
Reuse	`\input`, `\newcommand`, `\ensuremath`

Beginner tip. Do not try to memorize this table. The goal is recognition: when you need a little more space, a nonbreaking reference, an unnumbered line, or a reusable abbreviation, you should know that LaTeX probably has a command for it.

33 Horizontal Spacing in Math: From Tiny to Large

LaTeX already spaces mathematical operators well. Manual spacing is mainly for semantic separation: for example, before a differential, between a displayed equation and an indexing condition, or inside a

compact piece of notation.

What you type

```
$a\!b$ % negative thin space
$a\,b$ % thin space
$a\:b$ % medium space
$a\;b$ % thick space
$a\quad b$ % 1 em
$a\qquad b$ % 2 em
```

What appears in the PDF

$$ab \quad a\,b \quad a\,b \quad a\,b \quad a \quad b \quad a \quad b$$

The practical hierarchy is

$$\text{negative: } < \text{ small: } < \; < \; < \; < \; .$$

33.1 Research-paper examples

A differential often looks cleaner with a thin space:

What you type

```
\int_0^1 f(x)\,\mathrm{d}x
```

What appears in the PDF

$$\int_0^1 f(x)\,\mathrm{d}x$$

A condition can be separated from the main equation with `\qquad`:

What you type

```
y_i = \alpha + \beta x_i + \varepsilon_i,
\qquad i=1,\ldots,n.
```

What appears in the PDF

$$y_i = \alpha + \beta x_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

A negative thin space is sometimes useful when two symbols look too far apart:

What you type

```
\int\!\!\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y
```

What appears in the PDF

$$\iint f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

Common mistake. Do not insert `\quad` everywhere to make an equation "look nicer." Good mathematical spacing should usually communicate structure, not compensate for a poorly organized expression.

34 Fixed and Flexible Horizontal Space in Text

34.1 hspace: insert a specific amount of space

What you type

```
Left\hspace{2em}Right
```

What appears in the PDF

Left Right

Common units include `em` (roughly the width of the current letter M), `ex` (related to the current x-height), `pt`, `cm`, and fractions of text width such as `0.2\textwidth`.

Beginner tip. Inside prose, manual `\hspace` is rarely the right way to align material. Tables, minipages, or `\hfill` are usually more robust.

34.2 hfill: stretch space as much as possible

What you type

```
Draft version \hfill September 2026
```

What appears in the PDF

Draft version September 2026

This is useful when you want material on opposite sides of the same line.

What you type

```
John Maynard Keynes \hfill King's College
```

What appears in the PDF

John Maynard Keynes King's College

34.3 makebox: reserve a box of a chosen width

What you type

```
\makebox[3cm][l]{Coefficient:} 0.125
\makebox[3cm][l]{Std. error:} 0.041
```

What appears in the PDF

Coefficient: 0.125
Std. error: 0.041

The second optional argument controls alignment: `l`, `c`, or `r`.

Common mistake. For a real results table, use `tabular` or an imported table rather than manually aligning rows with `\makebox`. This command is mainly useful for small layout tasks.

35 Nonbreaking Spaces: The Tiny Tilde You Will Use Everywhere

In ordinary text, a space permits a line break. The character `~` creates a space but forbids a line break at that location.

What you type

```
Table~\ref{tab:main}
Figure~\ref{fig:event}
Section~\ref{sec:data}
Appendix~\ref{app:proofs}
Equation~\eqref{eq:baseline}
```

What appears in the PDF

For illustrative numbering, the PDF reads:

Table 1 Figure 2 Section 3 Appendix A Equation (1).

The tilde keeps each name and number together on one line; the actual numbers come from the labels.

This prevents awkward breaks such as

Table
3

and keeps “Table 3” together.

35.1 Other useful nonbreaking combinations

What you type

```
pp.~12--15
Figure~A1
$10~\mathrm{km}$
\$5~million
```

What appears in the PDF

pp. 12–15 Figure A1 10 km $5 million

Beginner tip. A simple habit: whenever a word and the number that follows it should stay together, consider `~`. This is especially useful for Table, Figure, Section, Appendix, Equation, Proposition, and page references.

36 Vertical Spacing: smallskip, medskip, bigskip, and vspace

For small manual adjustments, LaTeX provides named vertical skips:

What you type

```
First paragraph.

\smallskip
A little extra space.

\medskip
A medium amount of extra space.

\bigskip
A larger amount of extra space.
```

What appears in the PDF

First paragraph.

A little extra space.

A medium amount of extra space.

A larger amount of extra space.

For an exact amount, use `\vspace`:

What you type

```
\vspace{0.5em}
\vspace{6pt}
\vspace{0.25in}
```

What appears in the PDF

Used between visible material, these commands create increasingly explicit vertical gaps. For example:

Line above.

Line after `0.5em`.

Line after `6pt`.

Line after `0.25in`.

The starred form `\vspace*{...}` also works at the top of a page, where ordinary `\vspace` may disappear.

Common mistake. If you find yourself using many `\vspace` commands throughout a paper, the real problem is probably the document style, float placement, or a table/figure design. Manual vertical spacing should be occasional.

37 Paragraphs, Indentation, and Local Alignment

37.1 Start a new paragraph with a blank line

What you type

```
This is the first paragraph.

This is the second paragraph.
```

What appears in the PDF

This is the first paragraph.

This is the second paragraph.

A blank line is equivalent to `\par`:

What you type

```
This is the first paragraph.\par
This is the second paragraph.
```

What appears in the PDF

This is the first paragraph.

This is the second paragraph.

37.2 noindent and indent

What you type

```
\noindent This paragraph does not begin with an indent.

\indent This paragraph explicitly begins with an indent.
```

What appears in the PDF

This paragraph does not begin with an indent.

    This paragraph explicitly begins with an indent.

Because this handbook uses the `parskip` package, its paragraphs are separated by vertical space and normally have no indentation. Many journal styles instead use indentation and little or no paragraph spacing.

37.3 centering, raggedright, and raggedleft

Inside an environment such as a figure, `\centering` is usually preferable to opening another `center` environment.

What you type

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=.7\textwidth]{figures/main.pdf}
  \caption{Main result}
\end{figure}
```

What appears in the PDF

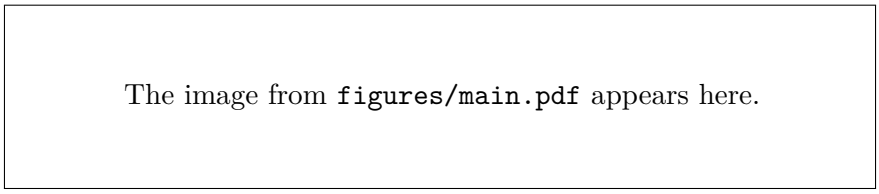


Figure 1: Main result

For local text alignment, group the declaration in braces:

What you type

```
{\raggedright Left-aligned text without full justification.\par}
{\centering Centered text.\par}
{\raggedleft Right-aligned text.\par}
```

What appears in the PDF

Left-aligned text without full justification.

Centered text.

Right-aligned text.

Beginner tip. Commands such as `\centering` are declarations: they remain active until the current group or environment ends. Braces are a convenient way to limit their scope.

38 Alignment Inside Equations: Ampersands, Line Breaks, notag, and intertext

In `align`, the ampersand `&` marks the alignment point. Many technical papers align at the equality sign.

What you type

```
\begin{align}
y_i &= \alpha + \beta x_i + \varepsilon_i, \\
```

```
\E[\varepsilon_i\mid x_i] &= 0.
\end{align}
```

What appears in the PDF

$$y_i = \alpha + \beta x_i + \varepsilon_i, \tag{6}$$
$$\mathbb{E}[\varepsilon_i \mid x_i] = 0. \tag{7}$$

Use `\notag` when one line should not receive an equation number:

What you type

```
\begin{align}
A &= B + C \notag\\
  &= D + E.
\end{align}
```

What appears in the PDF

$$\begin{aligned} A &= B + C \\ &= D + E. \end{aligned} \tag{8}$$

Use `\intertext{...}` when a sentence belongs between aligned lines:

What you type

```
\begin{align}
A &= B + C,\\
\intertext{Substituting the definition of $C$ gives}
A &= B + D + E.
\end{align}
```

What appears in the PDF

$$A = B + C, \tag{9}$$

Substituting the definition of C gives

$$A = B + D + E. \tag{10}$$

38.1 aligned: alignment inside another display

What you type

```
\[
\begin{aligned}
A &= B+C,\\
  &= D+E.
\end{aligned}
\]
```

What appears in the PDF

$$\begin{aligned} A &= B+C,\\ &= D+E. \end{aligned}$$

Beginner tip. Use `align` for a normal multi-line numbered display. Use `aligned` when the aligned block must live inside another math environment, such as inside brackets or one larger equation.

39 Invisible Space: phantom, hphantom, and vphantom

A phantom occupies the same space as its argument but prints nothing.

What you type

```
A = 1.25

\hphantom{A = }0.75
```

What appears in the PDF

A = 1.25
0.75

`\hphantom` reserves only horizontal space; `\vphantom` reserves only vertical space; `\phantom` reserves both.

A useful mathematical example is forcing delimiters to have matching height:

What you type

```
\left( x_i \vphantom{\frac{A}{B}} \right)
\qquad
\left( \frac{A}{B} \right)
```

What appears in the PDF

$$\left(x_i \vphantom{\frac{A}{B}} \right) \qquad \left(\frac{A}{B} \right)$$

Beginner tip. Phantoms are a fine finishing tool, but if you are using many of them to align a table, switch to a proper table-column specification instead.

40 Text Inside Math: text, mathrm, operatorname, and Friends

Mathematical letters are italic by default because LaTeX assumes they are variables. Words and named operators need different commands.

40.1 text: ordinary words inside an equation

What you type

```
y_i =
\begin{cases}
1, & \text{if employed},\\
0, & \text{otherwise}.
\end{cases}
```

What appears in the PDF

$$y_i = \begin{cases} 1, & \text{if employed}, \\ 0, & \text{otherwise}. \end{cases}$$

40.2 mathrm: upright mathematical labels

What you type

```
GDP_{it}^{\mathrm{real}}
SE(\hat\beta)
```

What appears in the PDF

$$GDP_{it}^{\mathrm{real}} \qquad SE(\hat{\beta})$$

40.3 operatorname: named mathematical operators

What you type

```
\operatorname{Cov}(X,Y)
\qquad
\operatorname{arg\,max}_{\theta} Q(\theta)
```

What appears in the PDF

$$\operatorname{Cov}(X,Y) \qquad \operatorname{arg\,max}_{\theta} Q(\theta)$$

For an operator you use repeatedly, define it once in the preamble:

What you type

```
\DeclareMathOperator{\plim}{plim}
\DeclareMathOperator*{\argmax}{arg\,max}
```

What appears in the PDF

These preamble declarations do not print anything by themselves. After they are defined, expressions can be written with operator-style spacing, for example

$$\operatorname*{plim}_{n\to\infty} \hat{\theta}_n = \theta_0 \qquad \arg\max_{\theta} Q(\theta).$$

Then write `\plim` or `\argmax` throughout the paper.

41 Math Alphabets: bold, blackboard, calligraphic, and Bold Greek

These commands encode different mathematical roles:

Source	Result	Typical use
`\mathbf{x}`	$\mathbf{x}$	Latin vector/matrix
`\bm{\beta}`	$\boldsymbol{\beta}$	bold Greek/vector
`\mathbb{R}`	$\mathbb{R}$	number sets
`\mathcal{F}`	$\mathcal{F}$	sets, sigma-fields, objective classes
`\mathrm{GDP}`	GDP	upright abbreviation in math

What you type

```
\mathbf{x}_i'\bm{\beta}
\qquad
\theta\in\mathbb{R}^K
\qquad
X\in\mathcal{F}
```

What appears in the PDF

$$\mathbf{x}_i'\boldsymbol{\beta} \qquad \theta \in \mathbb{R}^K \qquad X \in \mathcal{F}$$

Common mistake. Do not use bold merely for emphasis inside equations. A notation style should be consistent: for example, bold lowercase for vectors and bold uppercase for matrices.

42 Delimiter Size: left/right and Manual Alternatives

Ordinary parentheses do not grow automatically around tall material.

What you type

```
( \frac{A}{B} )
\qquad
\left( \frac{A}{B} \right)
```

What appears in the PDF

$$(\frac{A}{B}) \qquad \left(\frac{A}{B}\right)$$

The same works with brackets, braces, absolute values, and norms:

What you type

```
\left[ \frac{A}{B} \right]
\left\{ \frac{A}{B} \right\}
\left| x_i \right|
\left\| \mathbf{x} \right\|
```

What appears in the PDF

$$\left[\frac{A}{B}\right] \qquad \left\{\frac{A}{B}\right\} \qquad \left|x_i\right| \qquad \left\|\mathbf{x}\right\|.$$

Sometimes automatic sizing is too large. Manual sizes give finer control:

What you type

```
\bigl( x \bigr)
\Bigl( x \Bigr)
\biggl( x \biggr)
\Biggl( x \Biggr)
```

What appears in the PDF

$$(x) \qquad \bigl(x\bigr) \qquad \Bigl(x\Bigr) \qquad \Biggl(x\Biggr)$$

42.1 One-sided delimiters with a dot

What you type

```
\left. \frac{\partial f(x,\theta)}{\partial \theta}
\right|_{\theta=\theta_0}
```

What appears in the PDF

$$\left.\frac{\partial f(x,\theta)}{\partial \theta}\right|_{\theta=\theta_0}$$

The dot after `\left` means "no visible left delimiter."

43 Useful Annotation Commands: underbrace, overbrace, overset, and underset

These commands can make a derivation easier to read when used sparingly.

What you type

```
\underbrace{\alpha + \beta x_i}_{\text{systematic component}}
+ \varepsilon_i
```

What appears in the PDF

$$\underbrace{\alpha + \beta x_i}_{\text{systematic component}} + \varepsilon_i$$

What you type

```
\overbrace{x_1+\cdots+x_n}^{n\text{ observations}}
```

What appears in the PDF

$$\overbrace{x_1 + \cdots + x_n}^{n \text{ observations}}$$

Use `\overset` and `\underset` when you need a symbol directly above or below another relation:

What you type

```
\hat\theta_n \overset{p}{\longrightarrow} \theta_0
\qquad
x \underset{H_0}{\sim} F_0
```

What appears in the PDF

$$\hat\theta_n \overset{p}{\longrightarrow} \theta_0 \qquad x \underset{H_0}{\sim} F_0$$

44 substack: Put Multiple Conditions Under a Sum, Max, or Limit

A long subscript can be broken into multiple centered lines with `\substack`.

What you type

```
\sum_{\substack{i=1,\ldots,n\\ t=1,\ldots,T}} y_{it}
```

What appears in the PDF

$$\sum_{\substack{i=1,\ldots,n\\ t=1,\ldots,T}} y_{it}$$

Another common pattern is

What you type

```
\max_{\substack{c_1,c_2\\ c_1+c_2\le y}} u(c_1,c_2)
```

What appears in the PDF

$$\max_{\substack{c_1,c_2\\ c_1+c_2\le y}} u(c_1,c_2)$$

45 Dots: ldots, cdots, vdots, and ddots

Different kinds of dots are conventional in different contexts.

Command	Result	Typical use
`\ldots`	$\ldots$	comma-separated lists: $1, \ldots, n$
`\cdots`	$\cdots$	centered operations: $x_1 + \cdots + x_n$
`\vdots`	$\vdots$	vertical continuation in a matrix
`\ddots`	$\ddots$	diagonal continuation in a matrix

What you type

```
1,\ldots,n
\qquad
x_1+\cdots+x_n
```

What appears in the PDF

$$1, \ldots, n \qquad x_1 + \cdots + x_n$$

46 Relations and Definitions You Will See Often

You do not need to memorize all mathematical symbols, but several recur in quantitative papers:

Source	Result	Source	Result
`\approx`	$\approx$	`\sim`	$\sim$
`\propto`	$\propto$	`\equiv`	$\equiv$
`\neq`	$\neq$	`\leq`	$\leq$
`\geq`	$\geq$	`\in`	$\in$
`\subseteq`	$\subseteq$	`\perp`	$\perp$
`\coloneqq`	$\coloneqq$	`\implies`	$\implies$

What you type

```
\theta \in \Theta \subseteq \mathbb{R}^K,
\qquad
m(\theta) \coloneqq \E[g(W_i,\theta)].
```

What appears in the PDF

$$\theta \in \Theta \subseteq \mathbb{R}^K, \qquad m(\theta) \coloneqq \mathbb{E}[g(W_i, \theta)].$$

Beginner tip. `\coloneqq` is especially useful when you mean "is defined as," while `=` should usually mean equality. The command is supplied here by `mathtools`.

47 Reference Shortcuts Beyond ref and eqref

You already know `\ref` and `\eqref`. A few related commands are worth recognizing.

Command	What it does
`\ref{tab:main}`	Prints only the number, such as 3.
`\eqref{eq:model}`	Prints the equation number with parentheses, such as (2).
`\pageref{tab:main}`	Prints the page on which the label appears.
`\autoref{fig:main}`	With `hyperref`, prints the object type and number, such as "Figure 2."
`\nameref{sec:data}`	Prints the title of the labeled section, such as "Data."

What you type

```
See Table~\ref{tab:main} on page~\pageref{tab:main}.
The construction is described in \nameref{sec:data}.
```

What appears in the PDF

For example, if `tab:main` is Table 2 on page 12 and `sec:data` is the section titled "Data," the PDF reads:

> See Table 2 on page 12.
> The construction is described in Data.

The actual number, page, and section title come from the labels.

Beginner tip. For academic papers, explicit prose such as `Table~\ref{...}` is often clearer than relying on `\autoref` everywhere. But it is useful to know that automatic object names exist.

48 Links, Email Addresses, URLs, and Literal Paths

Because this handbook loads `hyperref`, several commands are available.

48.1 href: custom clickable text

What you type

```
\href{https://www.nber.org/}{NBER website}
```

What appears in the PDF

NBER website

48.2 url: print a URL safely

What you type

```
\url{https://www.example.com/data?id=10&year=2026}
```

What appears in the PDF

https://www.example.com/data?id=10&year=2026

48.3 mailto: clickable email

What you type

```
\href{mailto:name@university.edu}{name@university.edu}
```

What appears in the PDF

name@university.edu

48.4 nolinkurl: print URL-like text without a link

What you type

```
\nolinkurl{data/raw/county_panel_2026.csv}
```

What appears in the PDF

data/raw/county_panel_2026.csv

Beginner tip. `\url` and `\nolinkurl` are safer than manually escaping every underscore in a long URL or path.

49 Font Size Commands and Local Scope

LaTeX offers relative font-size declarations:

\tiny	\scriptsize	\footnotesize	\small	\normalsize	\large	\Large	\LARGE

Use braces to limit the effect:

What you type

```
Normal text.

{\small This paragraph is smaller, but only inside these braces.}

Normal text again.
```

What appears in the PDF

Normal text.

This paragraph is smaller, but only inside these braces.

Normal text again.

A common table pattern is

What you type

```
\begin{table}[htbp]
\centering
\small
\begin{tabular}{lcc}
...
\end{tabular}
\end{table}
```

What appears in the PDF

Variable	Mean	SD
Income	50.2	12.4

The `\small` declaration affects the table locally.

Common mistake. Do not shrink a table to unreadable size merely to force it onto one page. First shorten headings, remove unnecessary columns, split panels, or use landscape orientation.

50 Line and Page Control: Know the Difference

Command	Meaning
`\\`	End a line inside environments that expect line breaks; do not use it as a general paragraph command.
`\newline`	Force a line break in ordinary text. Rarely needed in papers.
`\linebreak`	Encourage/force a line break; optional strength `[0]`–`[4]`.
`\newpage`	Start a new page immediately, but queued floats may remain pending.
`\clearpage`	Print pending floats, then start a new page.
`\pagebreak`	Encourage/force a page break; optional strength `[0]`–`[4]`.
`\nopagebreak`	Discourage a page break at this point.
`\FloatBarrier`	From `placeins`; stop earlier floats from moving beyond this point.

For an appendix that should begin after all main-text figures and tables have appeared, a common pattern is

What you type

```
\clearpage
\appendix
\section{Additional Results}
```

What appears in the PDF

This code starts a new page after pending floats are placed, switches subsequent section labels to appendix letters, and then produces an appendix heading such as:

A Additional Results

51 Footnote Tricks: footnotemark and footnotetext

Usually you should write

What you type

```
The data come from the administrative records.\footnote{Access requires approval.}
```

What appears in the PDF

The data come from the administrative records.[1]

Sometimes the marker and the footnote text cannot be placed together, especially inside certain tables or captions. Then you may separate them:

What you type

```
Some text\footnotemark

...later...

\footnotetext{This is the corresponding footnote text.}
```

What appears in the PDF

Some text[1]

[1]This is the corresponding footnote text.

Common mistake. Footnotes inside tables and floats can be tricky. For regression-style or results tables, a table-note environment such as `threeparttable` is usually cleaner than ordinary footnotes.

52 Input, Include, and Includeonly

For a real research paper, `\input` is one of the most useful small commands.

What you type

```
\input{sections/introduction}
\input{sections/data}
\input{sections/results}
\input{tables/main_results}
```

What appears in the PDF

No filenames are printed. Instead, LaTeX inserts the contents of the four referenced `.tex` files at those four positions, in order.

[1]Access requires approval.

LaTeX inserts each file exactly where the command appears. The inserted file usually does *not* contain its own `\documentclass` or `document` environment.

`\include` is heavier: it starts the included material on a new page and creates separate auxiliary files. It is more common for books or dissertations than ordinary journal articles.

What you type

```
\include{chapter1}
\include{chapter2}
```

What appears in the PDF

No `\include` command is printed literally. The contents of `chapter1.tex` and `chapter2.tex` are inserted, with each included file beginning on a new page.

During a very large project, `\includeonly` can compile only selected included files:

What you type

```
\includeonly{chapter2}
```

What appears in the PDF

Nothing is printed by `\includeonly` itself. During compilation, only `chapter2.tex` is processed among the files named by `\include`.

Beginner tip. For a research paper, default to `\input`. It works naturally for sections, generated tables, appendix material, and reusable snippets.

53 newcommand and ensuremath: Make Your Own Shortcuts

If a long term or notation appears repeatedly, define it once.

What you type

```
% in the preamble
\newcommand{\outcome}{labor-force participation}

% in the paper
Our main outcome is \outcome.
```

What appears in the PDF

Our main outcome is labor-force participation.

For a notation command that should work both inside and outside math mode, `\ensuremath` is useful:

What you type

```
\newcommand{\ATE}{\ensuremath{\mathrm{ATE}}}

The parameter \ATE\ is reported in Table~\ref{tab:main}.
```

What appears in the PDF

The parameter ATE is reported in Table 1.

The exact table number is supplied by the referenced label.

For commands that take arguments:

What you type

```
\newcommand{\Econd}[2]{\E\!\left[#1\mid #2\right]}

\Econd{Y_i}{X_i}
```

What appears in the PDF

$$\mathbb{E}[Y_i \mid X_i]$$

Common mistake. Custom commands should improve consistency, not make your source unreadable. A collaborator should be able to infer what `\ATE` means; an opaque command such as `\zzzA` is not helpful.

54 A Few Useful Source-Code Controls

54.1 Percent: comment out the rest of a source line

What you type

```
This prints. % This does not print.
```

What appears in the PDF

This prints.

A percent sign can also suppress an unwanted source-code newline after a command definition or line break. Beginners do not need this trick often, but you may see it in templates:

What you type

```
\newcommand{\myname}{%
  John Maynard Keynes%
}
```

What appears in the PDF

The definition itself prints nothing. If you later type `\myname`, the visible result is:

John Maynard Keynes

The percent signs suppress unwanted source-line spaces in the command definition.

54.2 Temporarily hide a larger block

A simple package-free method is

What you type

```
\iffalse
This entire block is ignored by LaTeX.
It may contain several paragraphs.
\fi
```

What appears in the PDF

Nothing from the material between `\iffalse` and `\fi` appears in the PDF.

Common mistake. Do not leave large amounts of obsolete material hidden in a final project forever. Version control (for example, Git) is a better long-run archive than hundreds of commented-out lines.

55 Small Commands Worth Recognizing but Not Overusing

These are useful when you encounter them in coauthors' files or journal templates.

Command	Purpose
`\raisebox{1pt}{text}`	Move a small object vertically. Mostly a fine-tuning tool.
`\resizebox{\textwidth}{!}{...}`	Scale an object to a width. Useful, but often abused on tables.
`\scalebox{0.9}{...}`	Uniformly scale an object.
`\rotatebox{90}{...}`	Rotate text or a small object.
`\mbox{...}`	Put material in an unbreakable horizontal box.
`\fbox{...}`	Draw a visible frame around material; useful for debugging or draft callouts.
`\rule{width}{height}`	Draw a rule or create a precisely sized invisible/visible block.
`\strut`	Insert an invisible standard-height object; sometimes helps equalize row heights.

What you type

```
\rotatebox{90}{Outcome}
\qquad
\fbox{Draft result}
```

What appears in the PDF



56 Fast Lookup: "I Need to..."

I need to...	Try this first
add a small math space	`\,`, `\:`, or `\;`
add a larger math space	`\quad` or `\qquad`
tighten two math symbols slightly	`\!`
push text to the right edge	`\hfill`
add a precise horizontal gap	`\hspace{...}`
add a little vertical separation	`\smallskip`, `\medskip`, `\bigskip`
keep "Table 3" on one line	`Table~\ref{...}`
remove paragraph indentation	`\noindent`
center material in a figure/table	`\centering`
align equations at the equals sign	put `&` before `=` inside `align`
remove one equation number	`\notag`
insert prose between aligned equations	`\intertext{...}`
put ordinary words in math	`\text{...}`
write an upright label in math	`\mathrm{...}`
make Greek bold	`\bm{\beta}`
auto-size parentheses	`\left(...\right)`
label a component of an equation	`\underbrace{...}_{...}`
put two lines under a sum/max	`\substack{...\\...}`
write a sequence of values	`1,\ldots,n`
write a sum with centered dots	`x_1+\cdots+x_n`
refer to a page	`\pageref{...}`
print the name of a section	`\nameref{...}`
insert a clickable link	`\href{URL}{text}`
print a file path safely	`\nolinkurl{...}`
start appendix after all floats	`\clearpage` then `\appendix`
insert another .tex file	`\input{...}`
define a repeated abbreviation	`\newcommand`
keep a command usable in/out of math	`\ensuremath`
make a local font-size change	`{\small ...}`
stop floats crossing a boundary	`\FloatBarrier`

Beginner tip. This section is deliberately repetitive. A beginner learns these commands by seeing them repeatedly in different contexts, not by memorizing a vocabulary list once.

Part VI

Gap Check: More Small Commands Beginners Commonly Miss

This part is a second-pass audit of the handbook. The commands here are not exotic tricks. They are small pieces of LaTeX that experienced users reach for repeatedly, especially when polishing research papers, referee responses, appendices, and course notes.

57 A Command Name Can Eat the Space After It

A beginner often writes a command followed by ordinary text and is surprised that the space disappears.

What you type

```
\LaTeX is useful.
\LaTeX{} is useful.
\LaTeX\ is useful.
```

What appears in the PDF

LaTeXis useful.

LaTeX is useful.

LaTeX is useful.

What it means

A command made only of letters, such as `\LaTeX`, consumes the following source-space while TeX decides where the command name ends. The empty braces in `{}` end the command without printing anything. The control-space `\ ` explicitly inserts a normal space.

Beginner tip. For commands that produce text, `{}` is usually the cleanest way to preserve the following space: `\LaTeX{} paper`.

57.1 Abbreviations followed by a normal space

LaTeX treats a period after a lowercase or uppercase sequence as potentially sentence-ending. When the period is part of an abbreviation, a control-space can be useful:

What you type

```
Prof.\ Smith
U.S.\ data
Fig.\ 2
```

What appears in the PDF

Prof. Smith U.S. data Fig. 2

58 More Spacing Tools: enspace, hspace*, and vfill

The earlier spacing chapter covered the commands you will use most often. Three additional ones are worth recognizing.

Command	What it does
`\enspace`	Inserts a horizontal space roughly half an em wide.
`\hspace*{...}`	Like `\hspace`, but the space is preserved even at the beginning or end of a line.
`\vfill`	Inserts flexible vertical space that expands to fill the available height.

What you type

```
$A\enspace B$

\hspace*{2em}Indented with explicit space.
```

What appears in the PDF

$A\enspace B$

\hspace*{2em}Indented with explicit space.

A simple use of `\vfill` is to push material toward opposite ends of a fixed-height box:

What you type

```
\begin{minipage}[t][1.2in][t]{0.62\textwidth}
Top line
\vfill
Bottom line
\end{minipage}
```

What appears in the PDF

Top line

Bottom line

Common mistake. Do not use manual spacing to repair the overall layout of a paper. If many paragraphs require custom `\hspace` or `\vspace`, the underlying structure probably needs fixing.

59 Lists: itemize, enumerate, and description

Lists appear in theory assumptions, data descriptions, referee responses, research plans, and appendices.

59.1 Bullets with itemize

What you type

```
\begin{itemize}
  \item Administrative employment records
  \item County-level income data
  \item Firm characteristics
\end{itemize}
```

What appears in the PDF

- Administrative employment records
- County-level income data
- Firm characteristics

59.2 Numbered lists with enumerate

What you type

```
\begin{enumerate}
  \item Construct the analysis sample.
  \item Estimate the baseline specification.
  \item Report robustness checks.
\end{enumerate}
```

What appears in the PDF

1. Construct the analysis sample.
2. Estimate the baseline specification.
3. Report robustness checks.

Because this handbook loads `enumitem`, you can change the label style easily:

What you type

```
\begin{enumerate}[label=(\roman*)]
  \item relevance
  \item exclusion
  \item monotonicity
\end{enumerate}
```

What appears in the PDF

(i) relevance
(ii) exclusion
(iii) monotonicity

59.3 Term–definition lists with description

What you type

```
\begin{description}
  \item[Outcome.] Annual earnings.
  \item[Treatment.] Program eligibility.
  \item[Sample.] Workers aged 25--54.
\end{description}
```

What appears in the PDF

Outcome.
Annual earnings.
Treatment.
Program eligibility.
Sample.
Workers aged 25–54.

60 Fraction Size: frac, dfrac, tfrac, and displaystyle

The ordinary command `\frac` adapts to its context. Sometimes you want explicit control.

Command	Use
`\frac{a}{b}`	Normal context-sensitive fraction.
`\dfrac{a}{b}`	Force display-style fraction.
`\tfrac{a}{b}`	Force compact text-style fraction.
`\displaystyle`	Force display-style math for the following expression.

What you type

```
Inline: $\frac{1}{n}\sum_i x_i$

Larger inline: $\dfrac{1}{n}\sum_i x_i$

Compact: $\tfrac{1}{2}\sigma^2$

$\displaystyle \sum_{i=1}^{n} x_i$
```

What appears in the PDF

Inline: $\frac{1}{n}\sum_i x_i$

Larger inline: $\dfrac{1}{n}\sum_i x_i$

Compact: $\tfrac{1}{2}\sigma^2$

$$\displaystyle \sum_{i=1}^{n} x_i$$

Common mistake. Large display-style fractions can make a line of prose look uneven. Use `\dfrac` only when the larger fraction genuinely improves readability.

61 Absolute Values, Norms, Angles, and Conditional Bars

For mathematical delimiters, use semantic delimiter commands rather than typing keyboard characters blindly.

What you type

```
\lvert x \rvert
\lVert \beta \rVert
\langle x,y \rangle
\left\lvert \frac{x}{y} \right\rvert
\left\{x \middle| x>0\right\}
```

What appears in the PDF

$$\lvert x \rvert \qquad \lVert \beta \rVert \qquad \langle x, y \rangle \qquad \left\lvert \frac{x}{y} \right\rvert \qquad \left\{x \middle| x>0\right\}.$$

What it means

- `\lvert...\rvert`: absolute values.
- `\lVert...\rVert`: norms.
- `\langle...\rangle`: angle brackets or inner products.
- `\middle|`: a conditional bar whose height matches surrounding `\left` and `\right` delimiters.

62 Binomial Coefficients, Boxes, and Wider Accents

A few compact commands are useful in theory notes, proofs, and appendices.

What you type

```
\binom{n}{k}
\boxed{\beta>0}
\widehat{\beta_1+\beta_2}
\widetilde{Y}_{it}
\overline{X}_g
\vec{x}
\dot{x}_t
\ddot{x}_t
```

What appears in the PDF

$$\binom{n}{k} \qquad \boxed{\beta > 0} \qquad \widehat{\beta_1 + \beta_2} \qquad \widetilde{Y}_{it} \qquad \overline{X}_g \qquad \vec{x} \qquad \dot{x}_t \qquad \ddot{x}_t.$$

Beginner tip. `\widehat` and `\widetilde` are preferable to `\hat` and `\tilde` when the accent must span several symbols.

63 Two More Equation Environments: gather and multline

You already learned `equation`, `align`, `split`, and `aligned`. Two other `amsmath` environments solve different layout problems.

63.1 gather: centered equations with no alignment point

Use `gather` when several displayed equations belong together but do not need their equals signs aligned.

What you type

```
\begin{gather*}
E[u_i]=0,\\
\operatorname{Var}(u_i)=\sigma^2,\\
E[u_i x_i]=0.
\end{gather*}
```

What appears in the PDF

$$E[u_i] = 0,$$
$$\mathrm{Var}(u_i) = \sigma^2,$$
$$E[u_i x_i] = 0.$$

63.2 multline: one long equation over several lines

Use `multline` for one long equation that does not have a natural alignment point.

What you type

```
\begin{multline*}
Q(\theta)
= \sum_{i=1}^{n} q_i(\theta)
  + \lambda_1 P_1(\theta) \\
  + \lambda_2 P_2(\theta)
  + \lambda_3 P_3(\theta).
\end{multline*}
```

What appears in the PDF

$$Q(\theta) = \sum_{i=1}^{n} q_i(\theta) + \lambda_1 P_1(\theta) + \lambda_2 P_2(\theta) + \lambda_3 P_3(\theta).$$

63.3 tag: give an equation a custom visible tag

What you type

```
\begin{equation*}
Y_i = \alpha + \beta X_i + u_i.
\tag{A1}
\end{equation*}
```

What appears in the PDF

$$Y_i = \alpha + \beta X_i + u_i. \tag{A1}$$

Common mistake. Do not manually tag ordinary equations just to obtain the number you want. Use automatic numbering and `\label`/`\ref`. Manual tags are mainly useful when reproducing an external numbering convention.

64 Allowing a Break in a Long Mathematical Expression

A long inline expression may create an overfull line because TeX has too few legal break points. `\allowbreak` adds a possible break without forcing one.

What you type

```
$X_{it}=(age_{it},\allowbreak education_{it},
\allowbreak experience_{it},\allowbreak income_{it})$
```

What appears in the PDF

If the line fits, the PDF simply shows

$$X_{it} = (age_{it}, education_{it}, experience_{it}, income_{it}).$$

If space is tight, LaTeX is allowed to break the line at an `\allowbreak` location.

What it means

If the line fits, nothing visible happens. If the line is too long, LaTeX may break at an `\allowbreak`. This is useful for long lists of variables or parameters.

Common mistake. First ask whether the expression should be displayed on its own line. Do not sprinkle `\allowbreak` everywhere to hide poor equation design.

65 textwidth, linewidth, and columnwidth: Which Width Should You Use?

These length commands look similar but refer to different available widths.

Length	Meaning
`\textwidth`	Width of the main text block on the page.
`\linewidth`	Width currently available in the local environment. Inside a `minipage` or `subfigure`, this is usually the safest choice.
`\columnwidth`	Width of one column. In a one-column article it is usually the same as `\textwidth`; in a two-column document it is smaller.

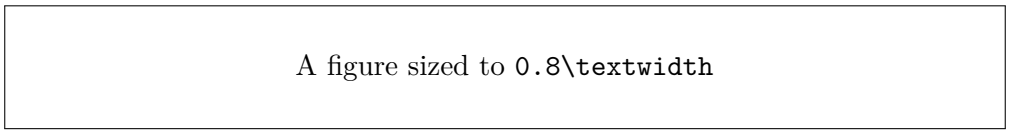
A common figure pattern is:

What you type

```
\begin{figure}
  \centering
  \includegraphics[width=0.8\textwidth]{main.pdf}
\end{figure}

\begin{subfigure}{0.48\textwidth}
  \includegraphics[width=\linewidth]{panel_a.pdf}
\end{subfigure}
```

What appears in the PDF



A panel sized to its local `\linewidth`

Beginner tip. Inside a `subfigure`, `minipage`, table cell, or other nested box, `\linewidth` often means what you actually want.

66 graphicspath: Stop Repeating the Figures Folder

If nearly all graphics live in the same folder, tell `graphicx` where to look.

What you type

```
\graphicspath{{figures/}{figures/appendix/}}

% Later:
\includegraphics[width=.8\textwidth]{main_result.pdf}
```

What appears in the PDF

The `\graphicspath` declaration prints nothing. Later, the `\includegraphics` command searches the listed folders and places `main_result.pdf` at 80% of the text width.

What it means

LaTeX first looks in the current directory, then in the listed graphics directories. The double braces are intentional: each path is wrapped in its own pair of braces.

Beginner tip. Many research projects still prefer explicit paths such as `figures/main_result.pdf` because they make the source easier to audit. Either convention is fine if the project is consistent.

67 Four Table Controls That Save a Lot of Frustration

67.1 tabcolsep: horizontal padding between columns

What you type

```
{\setlength{\tabcolsep}{4pt}
\begin{tabular}{lcc}
...
\end{tabular}}
```

What appears in the PDF

Variable	A	B
Treatment	0.12	0.03

The columns use 4 pt of horizontal padding.

The braces keep the change local to that table.

67.2 arraystretch: vertical row spacing

What you type

```
{\renewcommand{\arraystretch}{1.15}
\begin{tabular}{lcc}
...
\end{tabular}}
```

What appears in the PDF

Variable	A	B
Treatment	0.12	0.03
Control	0.08	0.01

The rows are slightly taller than the default.

A value larger than 1 makes rows taller.

67.3 p{width}: a column that wraps long text

What you type

```
\begin{tabular}{p{0.25\textwidth}p{0.65\textwidth}}
Variable & Definition \\
...
\end{tabular}
```

What appears in the PDF

Variable	Definition
Outcome	This longer definition wraps automatically inside the fixed-width paragraph column.

Unlike `l`, `c`, and `r`, a `p{...}` column wraps paragraphs automatically.

67.4 @{}: remove or customize edge padding

What you type

```
\begin{tabular}{@{}lcc@{}}
...
\end{tabular}
```

What appears in the PDF

Variable	A	B
Treatment	0.12	0.03

The empty `@{}` entries remove the normal outer padding.

The empty `@{}` at both ends removes the normal outer column padding. This is one reason professional tables often align more cleanly with surrounding text.

67.5 shortstack: a quick two-line column heading

What you type

```
\begin{tabular}{lcc}
\toprule
 & \shortstack{Annual\\earnings}
 & \shortstack{Employment\\rate} \\
\midrule
Treatment & 0.12 & 0.03 \\
\bottomrule
\end{tabular}
```

What appears in the PDF

	Annual earnings	Employment rate
Treatment	0.12	0.03

Common mistake. If headings are becoming very complicated, rethink the table structure before stacking three or four lines into every cell.

68 minipage and parbox: Small Independent Text Blocks

Both commands create a box with its own line width.

68.1 minipage: best for multi-line blocks

What you type

```
\begin{minipage}[t]{0.46\textwidth}
\textbf{Sample A}

Workers aged 25--40.
\end{minipage}
\hfill
\begin{minipage}[t]{0.46\textwidth}
\textbf{Sample B}

Workers aged 41--54.
\end{minipage}
```

What appears in the PDF

Sample A
Workers aged 25–40.

Sample B
Workers aged 41–54.

68.2 parbox: convenient for a smaller inline text box

What you type

```
\parbox{0.65\textwidth}{
This note wraps within a fixed-width box.
}
```

What appears in the PDF

This note wraps within a fixed-width box.

69 More natbib Citation Commands Worth Knowing

The first citation chapter covered `\citet` and `\citep`. These additional commands are useful when you need the pieces of a citation separately.

Command	Meaning	Typical output
`\citeauthor{key}`	Author name only	Author
`\citeyear{key}`	Year only	2020
`\citealp{key}`	Author-year without outer parentheses	Author, 2020
`\citealt{key}`	Textual-style citation without parentheses around year	Author 2020
`\nocite{key}`	Put an item in the bibliography without citing it in the text	no visible text here
`\nocite{*}`	Include every entry from the bibliography database	no visible text here

What you type

```
\citeauthor{acemoglu2024}
\citeyear{acemoglu2024}
\citealp{acemoglu2024}
\nocite{workingpaper2025}
```

What appears in the PDF

With a bibliography entry whose author is Acemoglu and year is 2024, the visible pieces are conceptually:

Acemoglu
2024
Acemoglu, 2024

The `\nocite` line prints no citation at that location; it only adds the entry to the bibliography.

Common mistake. Do not manually type author names and years merely to imitate a citation. If the statement is a citation, let the bibliography system generate it.

70 Page Numbering and Page Style Commands

Journal templates usually control these settings for you, but they are useful for working papers, appendices, referee responses, and course documents.

Command	Purpose
`\pagestyle{plain}`	Use a simple page style, usually page numbers.
`\pagestyle{empty}`	Suppress headers and footers on all following pages.
`\thispagestyle{empty}`	Suppress header/footer on this page only.
`\pagenumbering{roman}`	Use i, ii, iii, ... and reset the page counter.
`\pagenumbering{arabic}`	Use 1, 2, 3, ... and reset the page counter.
`\setcounter{page}{1}`	Set the current page counter manually.

What you type

```
\thispagestyle{empty}

\pagenumbering{roman}
% front matter

\clearpage
\pagenumbering{arabic}
% main paper
```

What appears in the PDF

These commands mainly affect page furniture rather than body text: the first page has no header/footer, front matter uses roman page numbers (*i*, *ii*, ...), and after the clear page the main text restarts with arabic numbers (1, 2, ...).

Common mistake. Do not override page styles in a journal or conference template unless the instructions explicitly allow it.

71 Unnumbered Sections That Still Appear in the Contents

The starred form removes automatic numbering:

What you type

```
\section*{Acknowledgments}
```

What appears in the PDF

Acknowledgments

The heading is unnumbered.

If you also want the unnumbered section in a table of contents:

What you type

```
\section*{Acknowledgments}
\addcontentsline{toc}{section}{Acknowledgments}
```

What appears in the PDF

Acknowledgments

The heading remains unnumbered, and "Acknowledgments" is also added to the table of contents.

What it means

The three arguments of `\addcontentsline` mean: write to the table of contents (`toc`), use section-level formatting (`section`), and print the text `Acknowledgments`.

Beginner tip. Most journal articles do not need a table of contents. This command is more relevant for dissertations, long reports, notes, and handbooks like this one.

72 Line Breaking and Hyphenation: Small Rescue Tools

72.1 newline versus a new paragraph

What you type

```
First line.\newline
Second line.

New paragraph after a blank source line.
```

What appears in the PDF

First line.
Second line.

New paragraph after a blank source line.

What it means

`\newline` starts a new line but does not start a new paragraph. In ordinary prose, a blank line is usually the correct way to begin a new paragraph.

72.2 A discretionary hyphen with \-

If LaTeX refuses to break a rare or technical word well, you can mark an allowed hyphenation point:

What you type

```
hetero\-geneity
counter\-factual
```

What appears in the PDF

Normally the words appear as *heterogeneity* and *counterfactual*. A hyphen is printed only if LaTeX actually breaks a line at the discretionary point.

The hyphen appears only if LaTeX actually breaks the word at that point.

72.3 Declare hyphenation for a whole document

What you type

```
\hyphenation{hetero-geneity counter-factual}
```

What appears in the PDF

Nothing is printed by this preamble command. It tells LaTeX that *heterogeneity* may break as hetero-geneity and *counterfactual* as counter-factual when line breaking requires it.

Put `\hyphenation` in the preamble.

72.4 emergencystretch: a last-resort paragraph safety valve

What you type

```
\setlength{\emergencystretch}{2em}
```

What appears in the PDF

Nothing is printed directly. LaTeX gains up to 2 em of extra stretch when it is trying to repair difficult paragraph line breaks.

This gives TeX a little extra flexibility when it is struggling to avoid overfull lines.

Common mistake. Do not use `\emergencystretch` as a substitute for fixing a genuinely unbreakable URL, a too-wide equation, or a bad table. It is a global typography adjustment.

73 One More Useful Distinction: center vs. centering

Both can center material, but they behave differently.

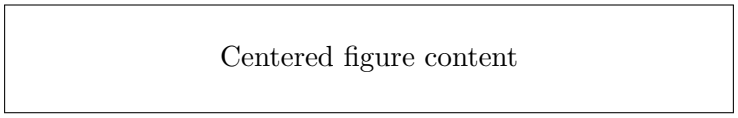
What you type

```
\begin{center}
Centered paragraph.
\end{center}

\begin{figure}
  \centering
  \includegraphics{figure.pdf}
\end{figure}
```

What appears in the PDF

Centered paragraph.



Both are centered; the `center` environment also adds vertical space around itself, whereas `\centering` is a declaration used inside the figure.

What it means

The `center` environment adds vertical space before and after itself. The declaration `\centering` changes alignment without adding that extra environment spacing. Inside `figure` and `table`, `\centering` is normally preferred.

74 A Second Fast Lookup: Commands Added After the Gap Check

I need to...	Try this
keep a space after a text-producing command	`{}` or `\`
insert a medium fixed horizontal space	`\enspace`
preserve horizontal space at a line edge	`\hspace*`
push material vertically apart	`\vfill`
make bullets	`itemize`
make numbered conditions	`enumerate`
make term–definition entries	`description`
force a large fraction	`\dfrac`
force a compact fraction	`\tfrac`
write an absolute value	`\lvert...\rvert`
write a norm	`\lVert...\rVert`
write an inner product	`\langle...\rangle`
write a binomial coefficient	`\binom`
span an accent over several symbols	`\widehat`, `\widetilde`
center several unrelated equations	`gather`
break one long displayed equation	`multline`
offer a legal break inside long inline math	`\allowbreak`
size a graphic to its local container	`\linewidth`
declare default figure folders	`\graphicspath`
tighten table columns	`\tabcolsep`
increase table row height	`\arraystretch`
wrap long text in a table column	`p{...}`
make a two-line table header quickly	`\shortstack`
put two text blocks side by side	`minipage`
print only a citation author	`\citeauthor`
print only a citation year	`\citeyear`
suppress the page style on one page	`\thispagestyle{empty}`
add an unnumbered section to the TOC	`\addcontentsline`
allow a manual word-break point	`\-`

Beginner tip. The goal is not to memorize this table. It gives you search vocabulary. When you know the name of the tool, you can quickly look up the exact syntax later.

Part VII

Showing Code and Referencing Things in Your Paper

75 Three Different Jobs: Inline Code, Code Blocks, and References

These three ideas look similar to a beginner, but they do different jobs:

What you want	Typical LaTeX tool
Show a command literally inside a sentence	`\verb`, `\lstinline`, or `\texttt`
Show several lines of source code	`verbatim` or `lstlisting`
Refer to a figure, table, equation, or section	`\label` + `\ref`, `\autoref`, or `\cref`

Beginner tip. When you want the PDF to literally show a command such as `\qquad`, you are *typesetting code*. When you write `Figure~\ref{fig:main}`, you are creating a *cross-reference*.

76 Inline Code: Showing a LaTeX Command Literally

Suppose you want your PDF to say: "Use `\qquad` for a large math-space." If you simply type `\qquad`, LaTeX tries to execute the command. To show the command itself, use a verbatim-style command.

76.1 The simplest tool: verb

What you type

```
Use \verb|\qquad| for a large horizontal space.
Use \verb|\textwidth| for the page text width.
```

What appears in the PDF

Use `\qquad` for a large horizontal space. Use `\textwidth` for the page text width.

What it means

The character immediately after `\verb` is a delimiter. Here the delimiter is `|`. LaTeX prints everything between the two delimiters literally.

You can choose another delimiter when the text itself contains `|`:

```
\verb+\qquad+
\verb!data_file.dta!
```

Common mistake. Do not write braces after `\verb`. It is `\verb|text|`, not `\verb{text}`. Also avoid `\verb` inside section titles and captions.

76.2 lstinline: inline code that matches your listings style

Because this handbook already uses the `listings` package, you can also write:

What you type

```
The command \lstinline|\qquad| inserts a large space.
The file \lstinline|analysis.R| contains the main estimation code.
```

What appears in the PDF

The command `\qquad` inserts a large space. The file `analysis.R` contains the main estimation code.

Beginner tip. For a conventional research paper, you usually do not need inline code often. It is most useful in notes, teaching material, appendices, replication documentation, and a LaTeX handbook like this one.

76.3 texttt: typewriter font, but not truly verbatim

What you type

```
\texttt{analysis.R}
\texttt{data\_clean.dta}
```

What appears in the PDF

`analysis.R` `data_clean.dta`

`\texttt` changes the font, but LaTeX still interprets reserved characters. That is why the underscore must be escaped in `data_clean.dta`. For arbitrary source code, `\verb` or `\lstinline` is easier.

77 Code Blocks: verbatim vs. lstlisting

77.1 verbatim: the basic built-in code block

The `verbatim` environment prints characters exactly as typed. Reserved characters such as `%`, `_`, `&`, and `\` do not need escaping inside it.

What you type

```
\begin{verbatim}
reg y x1 x2, robust
save results_main.dta, replace
\end{verbatim}
```

What appears in the PDF

```
reg y x1 x2, robust
save results_main.dta, replace
```

What it means

`verbatim` is simple and dependable. It is good when you only need to show literal text and do not care about captions, labels, frames, syntax highlighting, or line numbers.

77.2 lstlisting: better for polished code blocks

The `listings` package gives you much more control.

What you type

```
\begin{lstlisting}[language=R]
df <- read.csv("data/analysis.csv")
model <- lm(y ~ x1 + x2, data = df)
summary(model)
\end{lstlisting}
```

What appears in the PDF

```
df <- read.csv("data/analysis.csv")
model <- lm(y ~ x1 + x2, data = df)
summary(model)
```

Beginner tip. For Stata code, you can leave `language=` unspecified unless you define a custom Stata language. The code block will still print correctly.

78 Useful listings Options

A code block can have a caption, line numbers, a label, and a language.

What you type

```
\begin{lstlisting}[
  language=R,
  numbers=left,
  caption={Minimal data-cleaning example},
  label={lst:cleaning}
]
df <- subset(df, age >= 18)
df$log_income <- log(df$income)
\end{lstlisting}
```

What appears in the PDF

Listing 1: Minimal data-cleaning example

```
1 df <- subset(df, age >= 18)
2 df$log_income <- log(df$income)
```

Now you can refer to the listing just like a figure or table:

What you type

```
Listing~\ref{lst:cleaning-demo} shows the cleaning step.
```

What appears in the PDF

Listing 1 shows the cleaning step.

79 Include Code from an External File

In a research project, it is often better not to copy code into the paper manually. The `listings` package can read a source file directly.

What you type

```
\lstinputlisting[language=R]{code/main_analysis.R}
```

What appears in the PDF

The contents of `code/main_analysis.R` are inserted as a code listing. For example, if that file begins with two R commands, the PDF could show:

```
df <- read.csv("data.csv")
summary(df)
```

You can also include only part of a file:

```
\lstinputlisting[
  language=Python,
  firstline=20,
  lastline=45
]{code/figure1.py}
```

Beginner tip. This is useful for appendices, replication notes, teaching notes, or presentations. Most journal articles do not print large blocks of analysis code in the main paper itself.

80 The Basic Figure Reference Workflow

A figure has four jobs: create a float, insert the image, provide a caption, and attach a label.

What you type

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.75\linewidth]{figures/main_result.pdf}
  \caption{Main result}
  \label{fig:main-result}
\end{figure}
```

What appears in the PDF

Because this handbook does not ship the external image, an equivalent visible placeholder is:

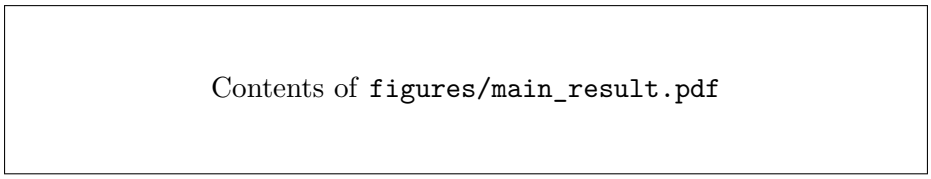


Figure 2: Main result

The most important ordering rule is:

caption first, label second

Then refer to it in your prose:

What you type

```
Figure~\ref{fig:main-result} reports the main result.
```

If the figure is numbered 2, the PDF automatically says:

What appears in the PDF

Figure 2 reports the main result.

Common mistake. Do not manually type "Figure 2" throughout your source. If the figure order changes, the number becomes wrong. Use `\ref`.

81 ref, eqref, autoref, cref, and Cref

There are several levels of convenience.

Command	Typical source	What it does
`\ref`	`Figure~\ref{fig:main}`	Prints only the number; you type "Figure" yourself.
`\eqref`	`\eqref{eq:main}`	Prints the equation number with parentheses.
`\autoref`	`\autoref{fig:main}`	`hyperref` automatically adds the object name.
`\cref`	`\cref{fig:main}`	`cleveref` automatically chooses "figure", "table", "section", etc.
`\Cref`	`\Cref{fig:main}`	Same as `\cref`, but capitalized for the start of a sentence.

For example, suppose a figure has `\label{fig:demo-ref}` and is numbered 3.

What you type

```
Figure~\ref{fig:demo-ref}
\autoref{fig:demo-ref}
\cref{fig:demo-ref}
\Cref{fig:demo-ref}
```

What appears in the PDF

If the referenced figure is number 3, these produce:

Figure 3
Figure 3
figure 3
Figure 3

Conceptually, these produce:

Figure 3
Figure 3
figure 3
Figure 3

Beginner tip. For beginners, `Figure~\ref{...}` is completely fine. Once your paper has many references, `cleveref` can reduce typing and prevent mistakes such as writing "Table" before a figure reference.

82 Referencing Several Objects at Once with cleveref

One advantage of `cleveref` is that it can format several references together.

What you type

```
\cref{fig:main,fig:robustness}
\cref{tab:summary,tab:main,tab:robust}
```

What appears in the PDF

Depending on the numbering, LaTeX can produce:

figures 2 and 3
tables 1–3

Depending on the numbering, LaTeX can automatically produce text such as:

figures 2 and 3
tables 1–3

You do not have to manually decide whether to write "Figure" or "Figures."

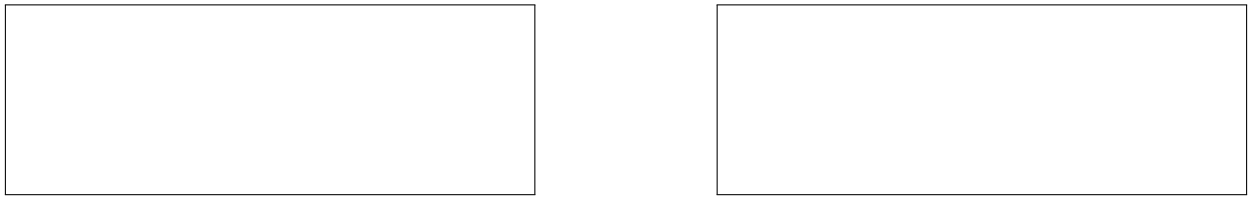
83 Panel References in Multi-Panel Figures

With the `subcaption` package, each panel can have its own label.

What you type

```
\begin{figure}[htbp]
  \centering
  \begin{subfigure}{0.47\textwidth}
    \includegraphics[width=\linewidth]{figures/panel_a.pdf}
    \caption{Full sample}
    \label{fig:panel-a-ref}
  \end{subfigure}
  \hfill
  \begin{subfigure}{0.47\textwidth}
    \includegraphics[width=\linewidth]{figures/panel_b.pdf}
    \caption{Restricted sample}
    \label{fig:panel-b-ref}
  \end{subfigure}
  \caption{Main results by sample}
  \label{fig:two-panel-ref}
\end{figure}
```

What appears in the PDF



(a) Full sample (b) Restricted sample

Figure 3: Main results by sample

Then you can write:

```
Figure~\ref{fig:two-panel-ref}\subref{fig:panel-a-ref}
Figure~\ref{fig:two-panel-ref}\subref{fig:panel-b-ref}
```

which gives forms such as **Figure 4(a)** and **Figure 4(b)**.

84 The Same Reference System Works for Tables, Equations, and Sections

What you type

```
\section{Data}
\label{sec:data-ref}

\begin{equation}
  y_i = \alpha + \beta x_i + \varepsilon_i.
  \label{eq:demo-ref}
\end{equation}

\begin{table}[htbp]
  \centering
  \caption{Summary statistics}
  \label{tab:summary-ref}
  ...
\end{table}
```

What appears in the PDF

The section heading, numbered equation, and table are created and labeled. Later references can therefore print forms such as:

> Section 2
> Equation (4)
> Table 3

The exact numbers depend on their positions in the document.

Then write:

```
Section~\ref{sec:data-ref}
Equation~\eqref{eq:demo-ref}
Table~\ref{tab:summary-ref}
```

The pattern is always the same:

`create object ⟶ label it ⟶ reference the label`

85 A Label-Naming Habit That Scales to a Dissertation

Use a short prefix that tells you what kind of object the label belongs to:

Prefix	Example
Section	`sec:data`
Equation	`eq:baseline`
Figure	`fig:event-study`
Table	`tab:main-results`
Appendix	`app:proofs`
Proposition	`prop:existence`
Assumption	`assump:sampling`
Code listing	`lst:cleaning`

This becomes especially valuable when your editor offers autocomplete after you type `\ref{`.

86 Common Reference Mistakes

Problem	What to check
Reference prints **??**	Compile again; then check that the label exists and is spelled identically.
Figure reference gives the wrong number	Put `\label` after `\caption`.
Two objects have the same reference number unexpectedly	Check for duplicated label names. Every label should be unique.
You typed Figure 3 manually and it became wrong	Replace hard-coded numbers with `\ref`.
`\ref` gives only "3", not "Figure 3"	That is normal. Type `Figure~\ref{...}` or use `\autoref`/`\cref`.
Citation or cross-reference is stale	Usually two clean recompilations are enough.

87 Fast Lookup: Code and References

I need to...	Use
Show `\qquad` literally in a sentence	`\verb|\qquad|` or `\lstinline|\qquad|`
Show a filename in typewriter font	`\texttt{analysis.R}`
Show several lines exactly as typed	`verbatim`
Show a polished code block	`lstlisting`
Number and reference a code block	`caption=` + `label=` in `lstlisting`
Read code directly from a file	`\lstinputlisting{...}`
Insert an image	`\includegraphics`
Give a figure a reference name	`\label{fig:...}` after `\caption`
Refer to the figure number	`Figure~\ref{fig:...}`
Automatically print the object type	`\autoref` or `\cref`
Refer to a displayed equation	`\eqref{eq:...}`
Refer to a figure panel	`\subref{fig:panel-a}`
Reference several objects together	`\cref{fig:a,fig:b}`

Part VIII

Tables: From First Table to Paper-Ready Table

88 The Difference Between table and tabular

This distinction confuses many beginners.

- `tabular` creates the rows and columns.
- `table` is a floating container that gives you a caption, label, and flexible placement.

89 Your First Table

What you type

```
\begin{tabular}{lcc}
Variable & Mean & SD \\
Income & 50.2 & 12.4 \\
Age & 39.7 & 11.3 \\
\end{tabular}
```

What appears in the PDF

Variable	Mean	SD
Income	50.2	12.4
Age	39.7	11.3

The column declaration `{lcc}` means:

- first column left aligned (`l`);
- second centered (`c`);

- third centered (`c`).

Every `&` moves to the next column. Every `\\` ends the row.

90 A Paper-Ready Table with booktabs

What you type

```
\begin{table}[htbp]
\centering
\caption{Summary Statistics}
\label{tab:summary}
\begin{tabular}{lccc}
\toprule
Variable & Mean & SD & N \\
\midrule
Outcome & 4.82 & 1.31 & 1,000 \\
Control & 2.14 & 0.87 & 1,000 \\
\bottomrule
\end{tabular}
\end{table}
```

What appears in the PDF

Table 1: Summary Statistics

Variable	Mean	SD	N
Outcome	4.82	1.31	1,000
Control	2.14	0.87	1,000

Use `\toprule`, `\midrule`, and `\bottomrule`; avoid a grid of vertical lines unless a journal/template specifically requires it.

91 Table Notes with threeparttable

What you type

```
\begin{table}[htbp]
\centering
\caption{Main Results}
\begin{threeparttable}
\begin{tabular}{lcc}
\toprule
& (1) & (2) \\
\midrule
Variable of interest & 0.125 & 0.110 \\
                     & (0.041) & (0.045) \\
\bottomrule
\end{tabular}
\begin{tablenotes}
\footnotesize
```

```
\item \textit{Notes:} Standard errors are in parentheses.
\end{tablenotes}
\end{threeparttable}
\end{table}
```

What appears in the PDF

Table 2: Main Results

	(1)	(2)
Variable of interest	0.125	0.110
	(0.041)	(0.045)

Notes: Standard errors are in parentheses.

92 Panels and cmidrule

What you type

```
\begin{tabular}{lcc}
\toprule
& (1) & (2) \\
\cmidrule(lr){2-3}
& Outcome A & Outcome B \\
\midrule
\multicolumn{3}{l}{\textit{Panel A: Full sample}} \\
Variable & 0.10 & 0.20 \\
\addlinespace
\multicolumn{3}{l}{\textit{Panel B: Subsample}} \\
Variable & 0.08 & 0.16 \\
\bottomrule
\end{tabular}
```

What appears in the PDF

	(1)	(2)
	Outcome A	Outcome B
Panel A: Full sample		
Variable	0.10	0.20
Panel B: Subsample		
Variable	0.08	0.16

93 What Does [htbp] Mean?

For `\begin{table}[htbp]` or `figure`, the letters suggest acceptable float positions:

- `h`: approximately here;
- `t`: top of a page;
- `b`: bottom of a page;

- `p`: dedicated float page.

LaTeX is allowed to move tables and figures to improve page layout.

Common mistake. Do not use `[H]` on every table and figure just because a float moved. Use it selectively. Too many forced floats can create ugly white space.

94 Common Table Errors

94.1 Wrong number of ampersands

If the table declares three columns, each row should normally contain two `&` separators.

```
\begin{tabular}{lcc}
Variable & Mean & SD \\ % correct: 3 columns
Income & 50.2 \\ % wrong: only 2 entries
\end{tabular}
```

The compiler may report **Extra alignment tab** or a confusing alignment error.

94.2 Forgetting row endings

```
Income & 50.2 & 12.4 \\
Age & 39.7 & 11.3 \\
```

Each completed row normally ends with `\\`.

94.3 A table is too wide

First, reduce unnecessary columns or shorten headings. If needed, use a smaller font locally:

```
\begin{table}[htbp]
\centering
\small
...
\end{table}
```

Avoid shrinking important tables until the text becomes unreadable.

Part IX

Figures and Graphics

95 Your First Figure

Load `graphicx` in the preamble. Then:

What you type

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.8\textwidth]{figures/main_result.pdf}
\caption{Main Result}
\label{fig:main}
\end{figure}
```

Because this handbook must compile without external files, the compiled example below uses a placeholder:

What appears in the PDF

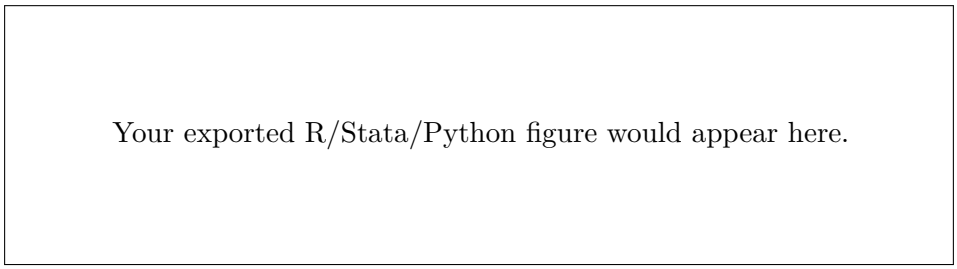


Figure 4: Main Result

96 Choosing Figure File Types

- **PDF**: excellent for vector plots exported from statistical software.
- **PNG**: good for screenshots and raster graphics.
- **JPG**: best for photographs, not usually statistical plots.

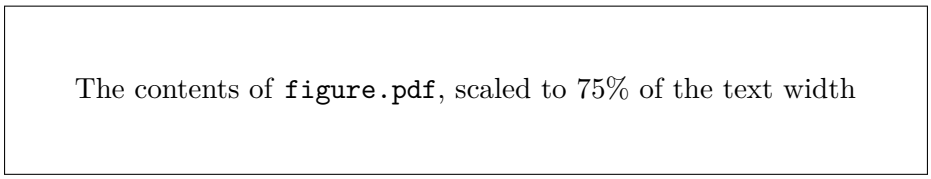
For line graphs and coefficient plots, vector PDF usually stays sharp when zoomed.

97 Sizing and Cropping Figures

What you type

```
\includegraphics[width=0.75\textwidth]{figure.pdf}
```

What appears in the PDF



Common alternatives:

```
\includegraphics[width=\textwidth]{figure.pdf}
\includegraphics[height=7cm]{figure.pdf}
\includegraphics[trim=10 10 10 10,clip,width=0.8\textwidth]{figure.pdf}
```

The four trim numbers mean left, bottom, right, top (in TeX points unless units are supplied).

98 Figure File Not Found

If LaTeX says it cannot find the image, check:

1. exact filename;
2. exact extension;
3. whether the file is in the expected folder;
4. capitalization (important on some systems);
5. whether the filename contains awkward characters.

A safe project layout is:

```
project/
  main.tex
  figures/
    fig1.pdf
    fig2.pdf
  tables/
  references.bib
```

99 Keeping Floats from Crossing a Section

If you want pending floats placed before the next section, load `placeins` and use:

```
\FloatBarrier
```

Use it selectively rather than after every paragraph.

Part X

Cross-References, Footnotes, Links, and Citations

100 Labels and References

Give important objects a label and let LaTeX insert the current number.

Object	Label convention	Refer with
Equation	`eq:baseline`	`Equation \eqref{eq:baseline}`
Table	`tab:main`	`Table \ref{tab:main}`
Figure	`fig:event`	`Figure \ref{fig:event}`
Section	`sec:data`	`Section \ref{sec:data}`
Appendix	`app:proofs`	`Appendix \ref{app:proofs}`

The prefixes are for humans, not LaTeX. They make labels easier to manage.

101 Where Should the label Go?

For figures and tables, put `\label` **after** `\caption`.

```
\caption{Main Results}
\label{tab:main}
```

For equations, put the label inside the equation environment.

Common mistake. If a table reference shows the section number instead of the table number, your `\label` may be before the caption.

102 Footnotes

What you type

```
This restriction is standard.\footnote{A longer explanation can go here.}
```

What appears in the PDF

This restriction is standard.[2]

LaTeX numbers footnotes automatically.

103 URLs and Hyperlinks

With `hyperref` loaded:

What you type

```
\url{https://www.nber.org/}

\href{https://www.nber.org/}{NBER website}
```

What appears in the PDF

https://www.nber.org/

NBER website

Use `\url{...}` for long URLs because it handles line breaking better than plain text.

104 Citations: natbib Basics

With `natbib`, the two commands you will use constantly are:

Command	Use	Typical output
`\citet{key}`	Author is part of your sentence	Author (2020)
`\citep{key}`	Entire citation is parenthetical	(Author, 2020)

[2] A longer explanation can go here.

What you type

```
\citet{sample2020} develops the framework.
A related literature studies this question \citep{sample2020}.
```

What appears in the PDF

With a sample bibliography entry "Author (2020)," the PDF would read:

> Author (2020) develops the framework.
> A related literature studies this question (Author, 2020).

The actual author names and year come from `references.bib`.

For this handbook, we will not execute those example keys, but in a real paper they refer to entries in `references.bib`.

105 Your references.bib File

A journal article entry might look like:

```
@article{sample2020,
  author = {Author, Alice and Scholar, Bob},
  title = {A Sample Economics Paper},
  journal = {Journal of Economic Examples},
  year = {2020},
  volume = {10},
  number = {2},
  pages = {100--125},
  doi = {10.xxxx/example}
}
```

At the end of your main `.tex` file:

```
\bibliographystyle{plainnat}
\bibliography{references}
```

Do not write `references.bib` inside `\bibliography{}`; write only the base name `references`.

106 Citation Variants You May Need

```
\citep{key1,key2} % multiple papers
\citep[see][]{key} % (see Author, 2020)
\citep[p.~12]{key} % page citation
\citet*{key} % full author list (style-dependent)
```

Common mistake. If citations show as question marks, check the citation key spelling, bibliography commands, and compile cycle before rewriting everything.

Part XI

Research Paper-Writing Patterns You Will Actually Reuse

107 Front Matter: Title, Affiliation, Contact, and Thanks

Research papers often need more information than a simple name under the title. A common pattern is to put the affiliation directly under the author name and use `\thanks` for contact information or acknowledgments.

What you type

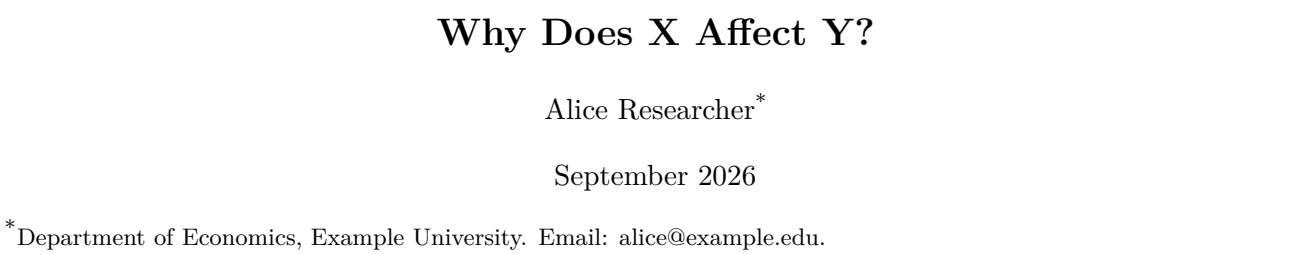
```
\title{Why Does X Affect Y?}

\author{
Alice Researcher\thanks{Department of Economics,
Example University. Email: alice@example.edu.}
}

\date{September 2026}

\begin{document}
\maketitle
\end{document}
```

What appears in the PDF



Why Does X Affect Y?

Alice Researcher*

September 2026

*Department of Economics, Example University. Email: alice@example.edu.

What it means

- `\thanks{...}` creates a title-page footnote.
- You can use it for affiliation, email, acknowledgments, funding, or a disclosure statement.
- For a working paper, keep the title block simple unless your department or conference has a required format.

Common mistake. Do not manually type an asterisk and then manually create a footnote. Let `\thanks` keep the marker and note connected.

108 Two or More Authors

For an early draft, the simplest approach is often enough:

What you type

```
\author{
Alice Researcher\\Example University
\and
Bob Economist\\Another University
}
```

What appears in the PDF

Alice Researcher
Example University

Bob Economist
Another University

The command `\and` tells the standard `article` class to separate authors. More elaborate journal title pages may require a journal template or a package, but beginners should first learn the standard mechanism.

109 Abstract, Keywords, and JEL Codes

A compact research-paper front matter block often includes an abstract followed by keywords or field-specific classification codes. The example below uses JEL codes from economics.

What you type

```
\begin{abstract}
This paper studies how X affects Y using newly assembled data.
The analysis documents three main findings and discusses their
implications for Z.
\end{abstract}

\noindent\textbf{Keywords:} labor markets, firms, productivity

\noindent\textbf{JEL codes:} J24, J31, O33
```

What appears in the PDF

Abstract

This paper studies how X affects Y using newly assembled data. The analysis documents three main findings and discusses their implications for Z.

Keywords: labor markets, firms, productivity

JEL codes: J24, J31, O33

Beginner tip. The exact JEL codes belong to your research topic. The LaTeX lesson here is simply how to typeset a compact metadata block.

110 Writing the Introduction with LaTeX, Not Fighting LaTeX

Your introduction is mostly ordinary prose. Do not put ordinary words into math mode, and do not add manual line breaks after every sentence.

What you type

```
\section{Introduction}
\label{sec:introduction}

How does X affect Y? This question matters because ...

This paper makes three contributions. First, ... Second, ...
Third, ...

The rest of the paper is organized as follows. Section~\ref{sec:data}
describes the data, Section~\ref{sec:design} explains the research
design, and Section~\ref{sec:results} presents the main results.
```

What appears in the PDF

1 Introduction

How does X affect Y? This question matters because ...

This paper makes three contributions. First, ... Second, ... Third, ...

The rest of the paper is organized as follows. Section 3 describes the data, Section 4 explains the research design, and Section 5 presents the main results.

Beginner tip. Use cross-references even while drafting. If sections move later, LaTeX updates the numbers automatically.

111 Defining Variables Clearly in Prose

Quantitative papers repeatedly introduce outcomes, predictors, treatments, controls, indices, and sample indicators. The cleanest habit is to define notation in a sentence immediately after it first appears.

What you type

```
Let $Y_{it}$ denote the outcome for unit $i$ in period $t$.
The variable $X_{it}$ is the main explanatory variable, and
$\mathbf{W}_{it}$ is a vector of observed controls.
The sample contains $N=12{,}480$ unit-period observations.
```

What appears in the PDF

Let Y_{it} denote the outcome for unit i in period t. The variable X_{it} is the main explanatory variable, and $\mathbf{W}_{it}$ is a vector of observed controls. The sample contains $N = 12{,}480$ unit-period observations.

What it means

- Use math mode only for symbols: `$Y_{it}$`, not for the surrounding sentence.
- `\mathbf{W}` makes a bold mathematical symbol for a vector.
- `12{,}480` protects the comma as punctuation rather than mathematical spacing.

112 A Small Notation Style Guide

Consistency matters more than any single notation choice. A common pattern is:

Meaning	Source	Output	Typical use
Scalar	`$x$`	x	one variable or parameter
Subscripted observation	`$x_i$`	x_i	observation i
Population parameter	`$\beta$`	β	unknown parameter
Estimate	`$\hat{\beta}$`	$\hat{\beta}$	estimated parameter
Vector	`$\mathbf{x}$`	$\mathbf{x}$	vector of variables
Expectation	`$\E[Y]$`	$\mathbb{E}[Y]$	expected value
Indicator	`$\mathbf{1}\{A\}$`	$\mathbf{1}\{A\}$	indicator for event A

Beginner tip. Pick one convention for vectors and estimates and use it everywhere. Inconsistent notation is much harder for a reader than simple notation.

113 Displaying a Baseline Specification Without Building a Formula Handbook

You will often need one or two displayed equations even in an empirical paper. The important LaTeX skill is the presentation: number the equation, label it, and define its objects in prose.

What you type

```
\begin{equation}
Y_{it} = \alpha + \beta X_{it} + \mathbf{W}_{it}'\gamma + u_{it}.
\label{eq:baseline}
\end{equation}

In Equation~\eqref{eq:baseline}, $Y_{it}$ is the outcome,
$X_{it}$ is the variable of interest, and $\mathbf{W}_{it}$
contains the controls. The coefficient $\beta$ is the parameter
of interest.
```

What appears in the PDF

$$Y_{it} = \alpha + \beta X_{it} + \mathbf{W}_{it}'\gamma + u_{it}. \tag{11}$$

In Equation (11), Y_{it} is the outcome, X_{it} is the variable of interest, and $\mathbf{W}_{it}$ contains the controls. The coefficient β is the parameter of interest.

Beginner tip. The purpose here is not to memorize a specific econometric model. It is to learn the paper-writing pattern: equation, label, reference, then definition.

114 Long Variable Names: Use Text or a Command

Do not force long English phrases into math italics.

What you type

```
% Hard to read
$average_income_{county,t}$
```

```
% Better for a named variable
$\text{Income}_{ct}$

% Or define a reusable command in the preamble
\newcommand{\Income}{\mathrm{Income}}
...
$\Income_{ct}$
```

What appears in the PDF

Better typesetting: Income_{ct} or Income_{ct}.

What it means

`\text{Income}` uses normal text inside math. `\mathrm{Income}` uses upright Roman letters and is useful for short variable names that behave like symbols.

115 Assumptions, Definitions, Propositions, and Proofs

Theory papers use these constantly, but empirical papers may also need formal assumptions or definitions. The `amsthm` package handles numbering and typography.

In the preamble:

```
\usepackage{amsthm}

\newtheorem{assumption}{Assumption}
\newtheorem{proposition}{Proposition}
\theoremstyle{definition}
\newtheorem{definition}{Definition}
```

Then in the paper:

What you type

```
\begin{assumption}[Sampling]
Observations are independently sampled from the population.
\end{assumption}

\begin{proposition}
Under Assumption~1, the object of interest is identified.
\end{proposition}

\begin{proof}
The argument follows in three steps. First, ... Second, ...
Finally, ...
\end{proof}
```

What appears in the PDF

Assumption 1 (Sampling)**.** *Observations are independently sampled from the population.*

Proposition 1. *Under Assumption 1, the object of interest is identified.*

Proof. The argument follows in three steps. First, ... Second, ... Finally, ... □

Common mistake. Do not manually type "Assumption 1" or "Proposition 2." Automatic numbering prevents errors when you add or delete results.

116 Numbered Conditions: (i), (ii), (iii)

Academic writing often lists assumptions, mechanisms, sample restrictions, or contributions using Roman numerals.

What you type

```
The analysis imposes three sample restrictions:
\begin{enumerate}[label=(\roman*)]
  \item observations must have nonmissing outcomes;
  \item the main covariate must be observed; and
  \item the unit must appear for at least two periods.
\end{enumerate}
```

What appears in the PDF

The analysis imposes three sample restrictions:

(i) observations must have nonmissing outcomes;
(ii) the main covariate must be observed; and
(iii) the unit must appear for at least two periods.

The `enumitem` package makes labels such as `(i)`, `(a)`, or `1.` easy to control.

117 Professional Summary-Statistics Tables

A useful summary-statistics table usually needs readable variable names, units when necessary, descriptive moments, and the number of observations.

What you type

```
\begin{table}[htbp]
\centering
\caption{Summary Statistics}
\label{tab:summary-econ}
\begin{threeparttable}
\begin{tabular}{lrrrrr}
\toprule
& Mean & SD & P25 & Median & N \\
\midrule
Outcome index & 0.42 & 0.18 & 0.29 & 0.41 & 12,480 \\
Annual income (USD) & 54,210 & 18,430 & 41,500 & 51,900 & 12,480 \\
Age (years) & 39.6 & 11.2 & 31 & 39 & 12,480 \\
\bottomrule
\end{tabular}
\begin{tablenotes}
\footnotesize
\item \textit{Notes:} The table reports descriptive statistics for
```

```
     the estimation sample. Monetary values are in 2026 dollars.
\end{tablenotes}
\end{threeparttable}
\end{table}
```

What appears in the PDF

Table 3: Summary Statistics

	Mean	SD	P25	Median	N
Outcome index	0.42	0.18	0.29	0.41	12,480
Annual income (USD)	54,210	18,430	41,500	51,900	12,480
Age (years)	39.6	11.2	31	39	12,480

Notes: The table reports descriptive statistics for the estimation sample. Monetary values are in 2026 dollars.

Beginner tip. The note should tell the reader what sample the table uses and clarify non-obvious units or transformations.

118 A Regression-Style Results Table

Even if Stata or R eventually generates your tables, it helps to understand the underlying LaTeX structure. Regression-style results tables commonly place estimates on one row and standard errors on the next, followed by specification information.

What you type

```
\begin{table}[htbp]
\centering
\caption{Main Results}
\label{tab:main-econ}
\begin{threeparttable}
\begin{tabular}{lccc}
\toprule
& (1) & (2) & (3) \\
& Outcome A & Outcome A & Outcome B \\
\midrule
Variable of interest & 0.118*** & 0.104*** & 0.072** \\
                   & (0.031) & (0.034) & (0.029) \\
\addlinespace
Controls & No & Yes & Yes \\
Unit fixed effects & No & Yes & Yes \\
Time fixed effects & No & Yes & Yes \\
Observations & 12,480 & 12,480 & 12,480 \\
$R^2$ & 0.18 & 0.31 & 0.27 \\
\bottomrule
\end{tabular}
\begin{tablenotes}
\footnotesize
\item \textit{Notes:} Standard errors are in parentheses.
     * $p<0.10$, ** $p<0.05$, *** $p<0.01$.
```

```
\end{tablenotes}
\end{threeparttable}
\end{table}
```

What appears in the PDF

Table 4: Main Results

	(1) Outcome A	(2) Outcome A	(3) Outcome B
Variable of interest	0.118***	0.104***	0.072**
	(0.031)	(0.034)	(0.029)
Controls	No	Yes	Yes
Unit fixed effects	No	Yes	Yes
Time fixed effects	No	Yes	Yes
Observations	12,480	12,480	12,480
R^2	0.18	0.31	0.27

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What it means

The first column is left aligned because it contains text. Result columns are centered here for simplicity. In a production table, packages such as `siunitx` can align numbers on decimal points.

119 Significance Stars Without Retyping Them

If you manually build a small table, a helper command can keep star formatting consistent.

In the preamble:

```
\newcommand{\sym}[1]{\ifmmode^{#1}\else\(^{#1}\)\fi}
```

Then:

What you type

```
0.118\sym{***}
```

What appears in the PDF

0.118^{***}

Beginner tip. If a statistical package generates the table automatically, let that package handle stars rather than adding them manually afterward.

120 Aligning Numbers on the Decimal Point

For tables with many numerical columns, `siunitx` can improve alignment.

In the preamble:

```
\usepackage{siunitx}
```

What you type

```
\begin{tabular}{l S[table-format=1.3] S[table-format=2.2]}
\toprule
Variable & {Estimate} & {Mean} \\
\midrule
A & 0.125 & 4.20 \\
B & -0.043 & 12.75 \\
\bottomrule
\end{tabular}
```

What appears in the PDF

Variable	Estimate	Mean
A	0.125	4.20
B	−0.043	12.75

The `S` column type understands numbers and aligns them at the decimal point.

121 Column Groups with multicolumn and cmidrule

Research tables often compare outcomes, samples, or specifications in grouped columns.

What you type

```
\begin{tabular}{lcccc}
\toprule
& \multicolumn{2}{c}{Full Sample}
& \multicolumn{2}{c}{Subsample} \\
\cmidrule(lr){2-3}\cmidrule(lr){4-5}
& (1) & (2) & (3) & (4) \\
\midrule
Variable & 0.10 & 0.12 & 0.08 & 0.09 \\
\bottomrule
\end{tabular}
```

What appears in the PDF

	Full Sample		Subsample	
	(1)	(2)	(3)	(4)
Variable	0.10	0.12	0.08	0.09

What it means

`\multicolumn{2}{c}{Full Sample}` makes one heading span two columns. `\cmidrule(lr){2-3}` draws a short rule under those two columns.

122 Importing a Table Generated by Stata or R

A productive research workflow often separates estimation from document formatting. Your statistical software writes a small `.tex` table body, and the paper imports it.

A project can look like:

```
paper/
  main.tex
  references.bib
  tables/
    main_results.tex
    summary_stats.tex
  figures/
    event_plot.pdf
```

Then your paper can use:

What you type

```
\begin{table}[htbp]
\centering
\caption{Main Results}
\label{tab:main-imported}
\input{tables/main_results.tex}
\end{table}
```

What appears in the PDF

The compiled PDF shows the contents of `tables/main_results.tex` exactly where `\input` appears, while `main.tex` stays short and readable.

Beginner tip. Decide whether your exported file contains only the `tabular` material or an entire `table` environment. Do not accidentally wrap a complete `table` inside another `table`.

123 A Figure with a Professional Note

Figures in research papers often need a note explaining the sample, construction, units, uncertainty, or confidence intervals.

What you type

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.78\textwidth]{figures/main_figure.pdf}
\caption{Outcome by Group}
\label{fig:main-econ}

\begin{minipage}{0.88\textwidth}
\footnotesize
```

```
\textit{Notes:} The figure plots group means by year.
Vertical bars denote 95\% confidence intervals.
\end{minipage}
\end{figure}
```

What appears in the PDF



Figure 5: Outcome by Group

Notes: The figure plots group means by year. Vertical bars denote 95% confidence intervals.

Beginner tip. A caption says what the figure is. A note tells the reader what is necessary to interpret it correctly.

124 Two-Panel Figures with subcaption

Multi-panel figures are common for heterogeneity, mechanisms, robustness, or different outcomes.

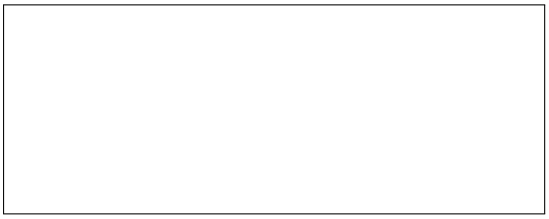
In the preamble:

```
\usepackage{subcaption}
```

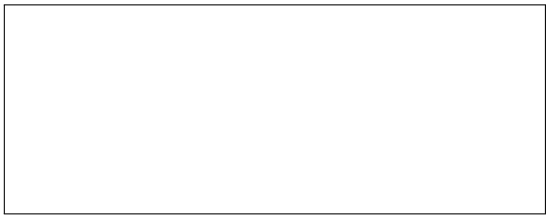
What you type

```
\begin{figure}[htbp]
\centering
\begin{subfigure}[t]{0.47\textwidth}
  \centering
  \includegraphics[width=\linewidth]{figures/panel_a.pdf}
  \caption{Outcome A}
  \label{fig:panel-a}
\end{subfigure}
\hfill
\begin{subfigure}[t]{0.47\textwidth}
  \centering
  \includegraphics[width=\linewidth]{figures/panel_b.pdf}
  \caption{Outcome B}
  \label{fig:panel-b}
\end{subfigure}
\caption{Effects Across Outcomes}
\label{fig:two-panel}
\end{figure}
```

What appears in the PDF



(a) Outcome A



(b) Outcome B

Figure 6: Effects Across Outcomes

What it means

Each panel is a `subfigure`. The entire object is still one numbered figure with one overall caption.

125 Referencing Panels, Tables, and Appendices in Sentences

Use the tilde `~` before a reference so the label and number do not split across lines.

What you type

```
Table~\ref{tab:main} reports the baseline estimates.
Figure~\ref{fig:main} shows the corresponding pattern visually.
Appendix~\ref{app:robustness} reports additional robustness checks.
Equation~\eqref{eq:baseline} defines the main specification.
```

What appears in the PDF

Table 2 reports the baseline estimates. Figure 3 shows the corresponding pattern visually. Appendix A reports additional robustness checks. Equation (1) defines the main specification.

Beginner tip. Never type "Table 2" manually in the source if the number can change.

126 A Variable-Definition or Data-Construction Table

For complicated data work, a compact variable table can be more readable than a long paragraph.

What you type

```
\begin{table}[htbp]
\centering
\caption{Key Variable Definitions}
\begin{tabularx}{0.95\textwidth}{@{}lXl@{}}
\toprule
Variable & Definition & Unit \\
\midrule
Outcome & Primary outcome constructed from administrative records. & Index \\
Income & Annual pre-tax labor income. & 2026 USD \\
Age & Age at the end of the calendar year. & Years \\
\bottomrule
```

```
\end{tabularx}
\end{table}
```

What appears in the PDF

Variable	Definition	Unit
Outcome	Primary outcome constructed from administrative records.	Index
Income	Annual pre-tax labor income.	2026 USD
Age	Age at the end of the calendar year.	Years

127 Showing Sample Construction as a Small Table

Sample construction is often clearer when the reader can see how many observations remain after each restriction.

What you type

```
\begin{tabular}{lr}
\toprule
Step & Observations \\
\midrule
Raw administrative records & 18,420 \\
Drop missing outcome & 16,905 \\
Apply age restriction & 14,212 \\
Require complete covariates & 12,480 \\
\bottomrule
\end{tabular}
```

What appears in the PDF

Step	Observations
Raw administrative records	18,420
Drop missing outcome	16,905
Apply age restriction	14,212
Require complete covariates	12,480

This is a good example of using LaTeX to improve exposition rather than adding more mathematics.

128 Robustness and Heterogeneity: Organize the Presentation

A robustness section often becomes hard to read because it contains many specifications. LaTeX helps if you organize the prose and references consistently.

What you type

```
\section{Robustness and Heterogeneity}
\label{sec:robustness}

\subsection{Alternative Samples}
Table~\ref{tab:robust-samples} repeats the analysis using ...
```

```
\subsection{Alternative Variable Definitions}
Table~\ref{tab:robust-definitions} shows that ...

\subsection{Heterogeneity}
Figure~\ref{fig:heterogeneity} reports estimates by ...
```

What appears in the PDF

6 Robustness and Heterogeneity

6.1 Alternative Samples Table 6 repeats the analysis using ...

6.2 Alternative Variable Definitions Table 7 shows that ...

6.3 Heterogeneity Figure 4 reports estimates by ...

Beginner tip. The LaTeX skill is simple sectioning and cross-referencing. The research judgment is deciding what belongs in the main text versus the appendix.

129 Appendix Table and Figure Numbering: A1, A2, ...

In the standard `article` class, `\appendix` changes section numbers to letters, but table and figure counters may continue from the main text. Many research papers instead use labels such as Table A1 and Figure A1.

What you type

```
\clearpage
\appendix

\setcounter{table}{0}
\renewcommand{\thetable}{A\arabic{table}}
\setcounter{figure}{0}
\renewcommand{\thefigure}{A\arabic{figure}}

\section{Additional Results}

\begin{table}[htbp]
\centering
\caption{Additional Results}
\label{tab:appendix-main}
...
\end{table}
```

What appears in the PDF

The first appendix table is numbered **Table A1**; the next is **Table A2**. The first appendix figure is **Figure A1**.

Common mistake. If your journal class already controls appendix numbering, follow the journal template instead of overriding it manually.

130 Appendix Numbering by Section: A.1, A.2, B.1, ...

For a long appendix, you may want table numbers to include the appendix section letter.

```
\appendix
\numberwithin{table}{section}
\numberwithin{figure}{section}
```

With appendix sections A and B, this can produce numbers such as Table A.1 and Table B.1.

Beginner tip. Choose one appendix numbering convention and keep it consistent across tables, figures, and references.

131 Wide Tables: First Fix the Content, Then the Layout

A wide table is common in empirical work. Do not immediately shrink it to unreadable text.

Try, in this order:

1. shorten column headings;
2. move details into the table note;
3. remove redundant columns;
4. use `\small` locally;
5. use a landscape page only if the table genuinely needs the width.

For a landscape page, add:

```
\usepackage{pdflscape}
```

Then:

```
\begin{landscape}
\begin{table}[p]
\centering
\caption{Wide Robustness Table}
...
\end{table}
\end{landscape}
```

The `pdflscape` package rotates the page for PDF viewing as well as printing.

132 Do Not Use resizebox as the First Solution

This compiles:

```
\resizebox{\textwidth}{!}{%
  \begin{tabular}{lcccccccc}
  ...
  \end{tabular}
}
```

But a table can become too small to read. Use `\resizebox` only after improving the table design itself.

133 Common Quantitative-Paper Typesetting Details

Small details appear constantly in drafts.

You want	Source	Output
five percent	`5\%`	5%
range of years	`2000-2010`	2000–2010
negative number	`$-0.25$`	-0.25
p-value	`$p<0.05$`	$p < 0.05$
R-squared	`$R\hat{2}$`	R^2
sample size	`$N=1{,}250$`	$N = 1{,}250$
dollars	`\$500`	$500
95 percent CI	`95\% CI`	95% CI
Table reference	`Table~\ref{tab:main}`	Table 2

Common mistake. A literal `$` starts or ends math mode. To print the currency symbol, type `\$`.

134 Percentage Points vs. Percent Signs

This is mostly a writing issue, but LaTeX makes the typography easy:

What you type

```
The outcome rises by 5\%.
The treatment increases the participation rate by 5 percentage points.
```

What appears in the PDF

The outcome rises by 5%. The treatment increases the participation rate by 5 percentage points.

Do not use the percent symbol when you mean percentage points simply because it is shorter to type.

135 Citing Papers in Different Sentence Positions

Citation commands should fit the grammar of the sentence.

What you type

```
\citet{author2020} studies a related mechanism.

A large literature studies this question \citep{author2020,author2022}.

The measurement follows \citet[p.~18]{author2020}.
```

What appears in the PDF

Author (2020) studies a related mechanism.

A large literature studies this question (Author, 2020; Author, 2022).

The measurement follows Author (2020, p. 18).

Beginner tip. Use `\citet` when the author is grammatically part of your sentence; use `\citep` when the citation itself is parenthetical.

136 Avoid Hard-Coding Names of Sections, Tables, and Equations

Bad drafting habit:

```
As shown in Table 4, ...
See Section 6 for robustness.
```

Better:

```
As shown in Table~\ref{tab:main}, ...
See Section~\ref{sec:robustness} for robustness.
```

If Table 4 becomes Table 5 later, your sentence remains correct automatically.

137 Useful Cross-Reference Naming Conventions

A predictable label system saves time in a long project.

Object	Example label	Meaning
Section	`sec:data`	data section
Equation	`eq:baseline`	baseline equation
Table	`tab:main`	main results table
Figure	`fig:heterogeneity`	heterogeneity figure
Appendix	`app:data`	data appendix
Proposition	`prop:main`	main proposition
Assumption	`assump:sampling`	sampling assumption

The prefix is only for you; LaTeX does not require it.

138 Reusable Commands for Long Names

If a phrase or notation appears repeatedly, define it once.

What you type

```
% Preamble
\newcommand{\mainoutcome}{employment rate}
\newcommand{\Y}{Y_{it}}

% Body
The main outcome is the \mainoutcome.
We denote it by $\Y$.
```

What appears in the PDF

The main outcome is the employment rate. We denote it by Y_{it}.

Beginner tip. Custom commands are most valuable when a notation or term may change later. One edit in the preamble can then update the entire paper.

139 Draft Notes That Cannot Accidentally Look Final

During a long paper project, visible draft markers are useful.

In the preamble:

```
\newcommand{\todo}[1]{\textcolor{red}{[TODO: #1]}}
```

Then:

What you type

```
The estimate remains stable across specifications.
\todo{Add appendix-table reference here.}
```

What appears in the PDF

The estimate remains stable across specifications. [TODO: Add appendix-table reference here.]

Common mistake. Search the source and final PDF for `TODO` before circulating or submitting the paper.

140 Anonymous and Named Drafts from One File

For conferences or journals that require anonymous submission, a simple switch can avoid maintaining two separate papers.

```
% In the preamble
\newif\ifanonymous
\anonymoustrue % switch to \anonymousfalse for named draft

\ifanonymous
  \author{Anonymous}
\else
  \author{Alice Researcher\\Example University}
\fi
```

This is more advanced than ordinary beginner syntax, but the idea is useful: keep one source of truth and switch only the information that must differ.

141 Double Spacing and Line Numbers for Submission

Some journals, workshops, or advisors request a referee-friendly draft.

In the preamble:

```
\usepackage{setspace}
\usepackage{lineno}
```

In the document:

```
\doublespacing
\linenumbers
```

For your normal working draft, you may prefer single spacing and no line numbers. Follow the actual submission instructions rather than assuming one format is universal.

142 A Reproducible Folder Structure for a Research Project

A clean project structure makes LaTeX easier to maintain and makes replication easier later.

```
project/
  paper/
    main.tex
    references.bib
    sections/
      introduction.tex
      data.tex
      design.tex
      results.tex
    tables/
      summary_stats.tex
      main_results.tex
    figures/
      main_figure.pdf
  code/
    01_clean_data.do
    02_analysis.do
  data/
    raw/
    derived/
```

Then `main.tex` can contain:

```
\input{sections/introduction}
\input{sections/data}
\input{sections/design}
\input{sections/results}
```

Beginner tip. LaTeX does not make a project reproducible by itself, but a predictable folder structure makes it much easier to connect code, tables, figures, and the written paper.

143 A More Realistic Research-Paper Skeleton

The following is not a rule for every paper. It is a useful beginner scaffold showing how the pieces in this handbook fit together.

```
\documentclass[11pt]{article}

\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,amsthm}
```

```
\usepackage{graphicx}
\usepackage{booktabs,threeparttable,tabularx}
\usepackage{subcaption}
\usepackage[round,authoryear]{natbib}
\usepackage[hidelinks]{hyperref}

\title{Paper Title}
\author{Your Name\\Department of Economics, Your University}
\date{\today}

\begin{document}
\maketitle

\begin{abstract}
Question, data or approach, main finding, and contribution.
\end{abstract}

\noindent\textbf{Keywords:} keyword 1, keyword 2, keyword 3

\noindent\textbf{JEL codes:} A00, B00

\section{Introduction}
\label{sec:introduction}

\section{Background or Institutional Setting}
\label{sec:background}

\section{Data}
\label{sec:data}

\section{Research Design}
\label{sec:design}

\begin{equation}
Y_{it}=\alpha+\beta X_{it}+\mathbf{W}_{it}'\gamma+u_{it}.
\label{eq:baseline-paper}
\end{equation}

Define every symbol in Equation~\eqref{eq:baseline-paper}.

\section{Results}
\label{sec:results}

Table~\ref{tab:main-paper} reports the main results.
Figure~\ref{fig:main-paper} provides a graphical summary.

\section{Robustness and Extensions}
\label{sec:robustness-paper}

\section{Conclusion}

\clearpage
\bibliographystyle{plainnat}
\bibliography{references}
```

```
\clearpage
\appendix
\section{Additional Results}
\label{app:additional}

\end{document}
```

What it means

This skeleton intentionally contains only one generic equation. The emphasis is on the writing system around the research: sections, labels, references, tables, figures, bibliography, and appendix.

Part XII

Appendices, Project Organization, and Reuse

144 Appendices

What you type

```
\clearpage
\appendix

\section{Additional Results}
\label{app:results}

\section{Additional Figures}
```

What appears in the PDF

After the page break, appendix mode changes the section labels to letters. The visible headings are conceptually:

A Additional Results

B Additional Figures

After `\appendix`, standard article sections are normally labeled with letters.

145 Page Breaks: newpage vs. clearpage

- `\newpage`: starts a new page.
- `\clearpage`: places pending figures/tables first, then starts a new page.

For starting an appendix or references section after many floats, `\clearpage` is often useful.

146 Splitting a Large Paper into Files

A dissertation chapter or long paper becomes easier to manage if sections are separated.

```
project/
  main.tex
  sections/
    introduction.tex
    data.tex
    methods.tex
    results.tex
  figures/
  tables/
  references.bib
```

In `main.tex`:

```
\begin{document}
\maketitle

\input{sections/introduction}
\input{sections/data}
\input{sections/methods}
\input{sections/results}

\end{document}
```

The section files usually contain only content, not another `\documentclass` or `\begin{document}`.

147 Custom Commands: Avoid Re-Typing Long Notation

Suppose you repeatedly write expectations and probabilities. Put in the preamble:

```
\newcommand{\E}{\mathbb{E}}
\newcommand{\Prob}{\mathbb{P}}
```

Then:

What you type

```
$\E[Y\mid X]$ and $\Prob(A)$
```

What appears in the PDF

$\mathbb{E}[Y \mid X]$ and $\mathbb{P}(A)$.

Beginner tip. Custom commands are valuable when notation appears many times. Do not create a macro for every one-off symbol.

148 Draft Comments and Temporary Notes

The simplest private note is a source comment:

```
% TODO: explain identification more clearly.
```

You can also define a visible draft command:

```
\newcommand{\todo}[1]{\textbf{[TODO: #1]}}
```

Then `\todo{rewrite this paragraph}` prints a visible reminder. Remove or disable such notes before submission.

149 Safe File Names

Prefer:

```
main.tex
references.bib
fig_event_study.pdf
tab_summary.tex
```

Avoid complicated punctuation and ambiguous file names. Spaces may work in modern systems, but simple names reduce cross-platform problems.

Part XIII

Common Mistakes and Debugging

150 The Beginner Error Map

Message or symptom	First thing to check
`Undefined control sequence`	Typo in a command, or missing package.
`Missing $ inserted`	Math syntax used outside math mode; unmatched dollar sign.
`Extra alignment tab &`	Too many ampersands in a table/alignment row.
`Missing } inserted`	Missing closing brace nearby.
`Runaway argument?`	Often a missing closing brace or environment end.
`Environment ... undefined`	Missing package or misspelled environment.
`... ended by \end{...}`	Begin/end environment names do not match.
Reference displays `??`	Compile again; verify the label exists and spelling matches.
Citation displays `?`	Verify citation key and bibliography workflow.
`File ... not found`	Wrong path, filename, extension, or capitalization.
`Overfull \hbox`	Something is too wide for the line/page.
Figure/table moved	Normal float behavior; placement is a request, not an absolute command.

151 Mistake: Missing or Extra Braces

Wrong:

```
\frac{a}{b
```

Correct:

```
\frac{a}{b}
```

When the error message looks strange, inspect the previous few lines. A missing brace can confuse LaTeX far beyond the original typo.

152 Mistake: Math Command Outside Math Mode

Wrong:

```
The coefficient is \beta.
```

Correct:

```
The coefficient is $\beta$.
```

153 Mistake: Unescaped Percent, Ampersand, or Underscore

Wrong prose:

```
The response rate is 80%.
R&D spending increases.
data_clean.csv
```

Correct prose:

```
The response rate is 80\%.
R\&D spending increases.
data\_clean.csv
```

154 Mistake: Mismatched Environments

Wrong:

```
\begin{align}
a &= b+c
\end{equation}
```

Correct:

```
\begin{align}
a &= b+c
\end{align}
```

155 Mistake: Using Double Dollars for Display Math

You may see old source files that use:

```
$$ y = a + bx $$
```

For modern LaTeX documents, prefer:

```
\[
y = a + bx
\]
```

or a numbered `equation` environment.

156 Mistake: Using Line-Break Commands for Paragraph Spacing

Wrong habit:

```
First paragraph.\\
\\
Second paragraph.
```

Better:

```
First paragraph.

Second paragraph.
```

Use `\\` mainly where a line break is structurally required, such as rows in tables or aligned equations.

157 Mistake: Manually Typing Numbers

Avoid:

```
As shown in Table 3 ...
Equation (5) implies ...
```

Prefer labels:

```
As shown in Table~\ref{tab:main} ...
Equation~\eqref{eq:model} implies ...
```

If you later insert a table earlier in the paper, the numbers update automatically.

158 Mistake: caption and label in the Wrong Order

Better:

```
\caption{Main Results}
\label{tab:main}
```

For tables and figures, `\label` normally comes after `\caption`.

159 Mistake: Forcing Every Float with [H]

`[H]` can be useful, but it says "put this float exactly here." If you force every table and figure, LaTeX may leave large blank areas.

Start with `[htbp]`. Use `[H]` only when you have a specific reason.

160 Mistake: Very Long URLs or Unbreakable Text

Use:

```
\url{https://example.com/a/very/long/path}
```

instead of pasting a long raw URL into prose. For code/file paths, `\path{...}` from `url/hyperref` can also help.

161 Overfull and Underfull Boxes

An **Overfull \hbox** warning means some line extends beyond its intended width. Common causes:

- a long URL;
- a wide equation;
- a table that is too wide;
- a long unbreakable filename or code string.

An **Underfull \hbox** warning means LaTeX had difficulty spacing a line attractively. It is often cosmetic, but you should visually inspect the page.

162 A Systematic Debugging Method

When something breaks:

1. Read the **first** meaningful error, not the 20 errors that follow it.
2. Inspect the reported line and several lines above it.
3. Check braces, dollar signs, `\begin`/`\end`, and special characters.
4. Ask what you changed most recently.
5. Comment out the suspicious block with `%` and compile again.
6. Restore pieces gradually until the error returns.

This "reduce the problem" strategy is much faster than randomly changing commands.

Part XIV

Complete Beginner Templates

163 Minimal Problem-Set Template

```
\documentclass[11pt]{article}

\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb}

\title{ECON 000: Problem Set 1}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

\section*{Question 1}

Write the explanation in prose first. Short notation can stay inline,
such as $\beta$ or $x_i$.

For an important displayed expression:
\[
y_i = \alpha + \beta x_i + \varepsilon_i.
\]

For multiple steps:
\begin{align*}
A &= B+C,\\
  &= D+E.
\end{align*}

\section*{Question 2}

Continue here.

\end{document}
```

164 Minimal Research Paper Template

```
\documentclass[11pt]{article}

\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb}
\usepackage{graphicx}
\usepackage{subcaption}
\usepackage{pdflscape}
\usepackage{booktabs,threeparttable}
\usepackage[round,authoryear]{natbib}
\usepackage[hidelinks]{hyperref}

\title{Paper Title}
\author{Your Name\\Department of Economics, Your University}
\date{\today}

\begin{document}
\maketitle
```

```
\begin{abstract}
One paragraph: question, approach, main finding, contribution.
\end{abstract}

\section{Introduction}
\label{sec:intro}

State the question, why it matters, what you do, what you find,
and how the paper contributes.

\section{Background}

\section{Data}
\label{sec:data}

\section{Research Design}

\begin{equation}
y_i = \alpha + \beta x_i + \varepsilon_i.
\label{eq:baseline}
\end{equation}

Define every object in Equation~\eqref{eq:baseline} in the text.

\section{Results}

Table~\ref{tab:main} reports the main result.

\begin{table}[htbp]
\centering
\caption{Main Results}
\label{tab:main}
\begin{threeparttable}
\begin{tabular}{lcc}
\toprule
& (1) & (2) \\
\midrule
Variable of interest & 0.125 & 0.110 \\
                     & (0.041) & (0.045) \\
\midrule
Observations & 1,000 & 1,000 \\
\bottomrule
\end{tabular}
\begin{tablenotes}
\footnotesize
\item \textit{Notes:} Explain the sample, standard errors,
and any notation needed to understand the table.
\end{tablenotes}
\end{threeparttable}
\end{table}

\section{Conclusion}

\clearpage
```

```
\appendix
\section{Additional Results}

\bibliographystyle{plainnat}
\bibliography{references}

\end{document}
```

165 Multi-File Paper Template

main.tex

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb}
\usepackage{graphicx,booktabs}
\usepackage[round,authoryear]{natbib}
\usepackage[hidelinks]{hyperref}

\title{Paper Title}
\author{Your Name}
\date{}

\begin{document}
\maketitle

\input{sections/abstract}
\input{sections/introduction}
\input{sections/data}
\input{sections/design}
\input{sections/results}
\input{sections/conclusion}

\clearpage
\bibliographystyle{plainnat}
\bibliography{references}
\end{document}
```

sections/introduction.tex

```
\section{Introduction}
\label{sec:intro}

Write the introduction here.
```

Part XV

Quick Reference for Daily Use

166 The Commands to Learn First

Command	What it does	Example result
`\section{...}`	section heading	numbered heading
`\textbf{...}`	bold text	**bold**
`$...$`	inline math	x_i
`\[...\]`	display math	centered equation
`_`	subscript	x_i
`^`	superscript	x^2
`\frac{a}{b}`	fraction	$\frac{a}{b}$
`\sum`	summation	$\sum_i x_i$
`\begin{align*}`	aligned derivation	aligned lines
`\text{...}`	words inside math	$x = 1$ if treated
`\begin{table}`	table float	numbered table
`\begin{tabular}`	rows/columns	table body
`\includegraphics`	image import	figure
`\caption`	table/figure title	caption
`\label`	target for reference	internal label
`\ref`	table/figure/section reference	number
`\eqref`	equation reference	(number)
`\footnote`	footnote	footnote marker
`\citet`	textual citation	Author (year)
`\citep`	parenthetical citation	(Author, year)
`\multicolumn`	grouped table heading	spans columns
`\input`	import another .tex file	inserts file contents
`\appendix`	begin appendix	lettered appendix sections
`\newcommand`	reusable notation/text	custom command

167 Spacing, Alignment, and Tiny Commands - Fast Lookup

These are the small commands you will use constantly while writing. For most of them, you do not need to memorize the exact spacing size; remember the rough order and copy the command when needed.

Math spacing ladder

Command	What it does	Compiled example
`\,`	tiny positive math space	$a\,b$
`\:`	small/medium math space	$a\:b$
`\;`	medium-large math space	$a\;b$
`\!`	small *negative* math space	$a\!b$
`\quad`	large fixed space (about 1 em)	$a \quad b$
`\qquad`	extra-large fixed space (about 2 em)	$a \qquad b$

A useful visual order is

$$a\!b \quad < \quad a\,b \quad < \quad a\:b \quad < \quad a\;b \quad < \quad a \quad b \quad < \quad a \qquad b.$$

A common quantitative-paper pattern is to separate an equation from an index or condition:

```
Y_i = \alpha + \beta X_i + \varepsilon_i,
\qquad i=1,\ldots,n.
```

which compiles as

$$Y_i = \alpha + \beta X_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Beginner tip. Use `\quad` or `\qquad` for visual separation, not to fake equation alignment. If several equations should line up at their equals signs, use `align` with `&`.

Alignment and layout shortcuts

Command	Use	Typical pattern
`&`	alignment point in `align` or column separator in tables	`Y_i &= a+bX_i`
`\\`	end a row/line in `align`, `tabular`, etc.	`... \\`
`\notag`	suppress one equation number in `align`	`... \notag`
`~`	nonbreaking space	`Table~\ref{tab:main}`
`\hfill`	flexible horizontal space; pushes material apart	`Left \hfill Right`
`\hspace{...}`	exact horizontal space	`\hspace{1em}`
`\smallskip`	small vertical gap	between nearby blocks
`\medskip`	medium vertical gap	between nearby blocks
`\bigskip`	larger vertical gap	between nearby blocks
`\vspace{...}`	exact vertical space	`\vspace{0.5em}`
`\noindent`	suppress paragraph indentation once	`\noindent Text...`
`\centering`	center material inside a figure/table/group	before `\includegraphics`
`\phantom{...}`	reserve invisible width/height	manual alignment trick

For aligned equations, remember this pattern:

```
\begin{align}
Y_i &= \alpha + \beta X_i + \varepsilon_i, \\
E[\varepsilon_i\mid X_i] &= 0. \notag
\end{align}
```

The `&` marks the alignment point; placing it immediately before `=` lines up the equals signs.

Tiny math-writing helpers worth remembering

Command	Why it is useful	Example
`\text{...}`	normal words inside math	$D_i = 1$ if treated
`\mathrm{...}`	upright math letters/units	$10\,\mathrm{km}$
`\operatorname{Var}`	custom operator with proper spacing	$\operatorname{Var}(X)$
`\left(...\right)`	automatic delimiter sizing	$\left(\frac{a}{b}\right)$
`\dfrac{a}{b}`	display-style fraction in inline math	$\dfrac{a}{b}$
`\lvert x\rvert`	properly sized absolute value	$\lvert x \rvert$
`\lVert x\rVert`	norm	$\lVert x \rVert$
`\underbrace`	annotate part of an expression	$\underbrace{x_1 + x_2}_{\text{two terms}}$
`\substack`	stack conditions under sums/limits	$\sum_{\substack{i=1 \\ i \neq j}}^{n} x_i$
`\ldots` / `\cdots`	ellipses in lists / centered math	$x_1, \ldots, x_n$; $A_1 \cdots A_n$

Show a command literally in your PDF

If you want the PDF to display a command rather than execute it, use inline verbatim:

```
Use \verb|\qquad| for a large horizontal space.
```

This prints the command `\qquad` literally. With the `listings` package, `\lstinline|...|` is another convenient option.

Fast "I need to..." lookup

I need to...	Try this first
add a tiny math space	`\,`
add a medium math space	`\;`
add a large math space	`\quad`
add an even larger math space	`\qquad`
slightly tighten math spacing	`\!`
keep "Table 3" on one line	`Table~\ref{...}`
push one item to the far right	`\hfill`
line up several equals signs	`align` + `&=`
remove one equation number	`\notag`
put ordinary words in math	`\text{...}`
make parentheses scale to a fraction	`\left( ... \right)`
add a small vertical gap	`\smallskip`
remove one paragraph indent	`\noindent`
show `\qquad` literally	`\verb|\qquad|`

168 Reserved Characters - Fast Lookup

Want	%	$	&	_	#	{	}	~	\
Type	`\%`	`\$`	`\&`	`\_`	`\#`	`\{`	`\}`	`\textasciitilde`	`\textbackslash`

169 Before You Submit: Ten Checks

1. Compile from a clean state and check for real errors.
2. Search the PDF for **??** and unresolved citations.
3. Check that every figure/table is mentioned in the text.
4. Check that every figure/table has a useful caption.
5. Check table notes for sample, standard errors, units, and abbreviations where relevant.
6. Make sure equation symbols are defined near first use.
7. Inspect wide equations and tables for margin overflow.
8. Remove TODO notes and draft comments that should not be public.
9. Verify the bibliography style and all cited references.
10. Open the final PDF separately and visually inspect several pages, especially tables and figures.

170 A First-Week Learning Plan

1. **Day 1:** compile the minimal document; learn text, sections, and paragraphs.
2. **Day 2:** learn inline/display math, subscripts, superscripts, fractions, and Greek letters.
3. **Day 3:** learn `equation`, `align`, labels, and references.

4. **Day 4:** create one clean table and insert one figure.
5. **Day 5:** create a `references.bib` file and practice `\citet`/`\citep`.
6. **Day 6:** split a small paper into files with `\input`.
7. **Day 7:** intentionally create and fix several errors from the debugging section.

171 Final Advice

For an academic writer, the goal is not to become a LaTeX programmer. The useful skill is to become fast at turning research content into a clean, reproducible document.

A reliable workflow is:

1. start from a working template;
2. make one change at a time;
3. compile frequently;
4. reuse labels, tables, figures, and commands consistently;
5. let LaTeX handle numbering and layout whenever possible;
6. debug systematically instead of randomly changing syntax;
7. keep a short cheat sheet open while writing.

Most day-to-day academic writing uses a surprisingly small subset of LaTeX. For research writing, the recurring tasks are especially predictable: clean prose, notation, cross-references, results tables, figures, citations, appendices, and imported outputs from statistical software. Once the commands in this handbook become familiar, you can learn specialized packages only when a real research need appears.